

A Symmetry-Quotient Reduction for Odd-Characteristic SNOVA: Exact Structure and Conditional Recovery Ledgers

Justin Thaler*

Kai Zhang[†]

Scott Kominers[‡]

Quang Dao[§]

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Abstract

SNOVA is a multivariate signature scheme that compresses the public key of Unbalanced Oil and Vinegar by representing variables and coefficients with small matrices. After wedge attacks weakened its characteristic-two design, SNOVA adopted odd-prime fields and symmetric public matrices. This paper shows that the symmetry creates a different weakness in the public verifier.

We analyze the common-column relation already used in affine-column forgery attacks. For a symmetric odd-characteristic SNOVA instance whose public matrix algebra has dimension d , symmetry reduces the number of distinct restricted quadratic forms from d^2 to $d(d+1)/2$. Thus sixteen forms become ten when $\ell = 4$, and four become three when $\ell = 2$. An invertible public change of output coordinates then separates these quadratics from the remaining affine constraints and target checks. For every published Version 2.3 parameter shape that passes the stated public rank checks, this gives an exact quadratic residual system on a public affine space. The reduction permits affine slices of any dimension and preserves every verifier equation, rejection rule, and final verification check.

The recovery estimates use Polynomial XL. For the three $\ell = 2$ shapes, we prove full affine coefficient support on transverse slices. For $\ell = 4$, we prove an exact graph-complement realization of all ten homogeneous channels on a zero-offset slice. Positive-rank offsets require a separate publicly checkable compatibility premise whose frequency under official key generation is unproved. Under the ordinary PXL assumptions and an instancewise extraction premise, the support-constrained route lies below the Level III and V reference exponents but not Level I. A second, manuscript-specific structured PXL premise gives conditional estimates below the reference exponents for all nine shapes. The lowest unit-leading gate exponents for $\omega = 2.81$ are 128.833, 187.030, and 245.042 for $\ell = 2$, and 134.038, 182.006, and 230.444 for $\ell = 4$. These figures omit auxiliary work and are not wall-clock or peak-memory bounds. We do not claim a production-size solve or forgery.

1 Introduction

NIST’s Additional Digital Signatures process seeks signature schemes with security and performance profiles that complement the standards selected in its first post-quantum process. SNOVA is one of the nine candidates advanced to the third round in May 2026 [1, 2]. It belongs to the Unbalanced Oil and Vinegar line of multivariate signatures, where verification evaluates a public quadratic map and signing uses a hidden central map with an oil-and-vinegar structure [3, 4].

*a16z crypto research and Georgetown University

[†]MIT

[‡]Harvard University and a16z crypto research

[§]Carnegie Mellon University and LayerZero Labs

SNOVA obtains compact public keys by representing variables and coefficients with small matrices and generating many public forms from a smaller structured family. That structure has led to key-recovery and direct forgery attacks. The former expose the hidden UOV structure, while Beullens uses a public relation among signature columns. Later work improves both lines through group actions, extension-field lifting, and multihomogeneous solving [5–10]. These works analyze the original or Round 2 construction over $\mathbb{F}_{16}$.

The next revision answered Ran’s wedge attack, which uses the alternating polar forms available in characteristic two. Bros et al. adapt the method to SNOVA’s block structure [11, 12]. The SNOVA team responded with odd-prime fields and symmetric public quadratic matrices, while warning that symmetrization could create new attacks [13]. NIST later described this change as a response to the wedge attack and did not regard SNOVA as stable [2].

This change entered the specification in Version 2.1, not Version 2.3. Version 2.3 subsequently added rectangular signatures and revised parameter sets [4]. Our quotient theorem concerns the symmetric odd-characteristic construction and is not specific to those nine parameter sets. The exact rank and dimension analysis covers all nine published $q = 19$ Version 2.3 shapes. The audited numerical PXL ledger covers both extension degrees, with separate premises for the support-constrained and structured $\ell = 4$ routes.

Prior attacks already cover odd characteristic. Jin et al. give a UOV key-recovery attack in arbitrary characteristic [14]. Ran gives an odd-characteristic construction using symmetric forms, but SNOVA’s added structure requires separate analysis [11]. Ding, Li, and Tao give a wedge framework valid in any characteristic, while their concrete SNOVA estimates cover the even-characteristic Round 2 parameters [15]. None combines the symmetric public matrices in the current $q = 19$ construction with the affine-column forgery relation.

This brings us to the question addressed in this paper. What happens when an affine-column attacker imposes the same common-column relation on the new symmetric verifier? We prove that symmetry makes pairs of restricted quadratic forms identical. An invertible public change of output coordinates then separates these quadratics from the remaining affine constraints and target checks. For every published $q = 19$ parameter shape that passes the stated public rank checks, this leaves a quadratic system on a public affine space. The attacker may choose a square, overdetermined, or underdetermined affine slice without dropping a verifier equation.

This statement concerns the restricted verifier, not the unrestricted matrix-valued map. Let $A = \mathbb{F}_q[S]$ have dimension d , and let $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$. After all signature columns are expressed through one stacked column u , symmetry gives $q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u)$. The d^2 ordered labels therefore factor through $\text{Sym}^2(A)$, with the exact dimension change

$$d^2 \longrightarrow \binom{d+1}{2}, \quad 16 \longrightarrow 10 \text{ for } d = 4, \quad 4 \longrightarrow 3 \text{ for } d = 2. \quad (1)$$

The full verifier relates the two ordered features by matrix transposition; only the common-column restriction makes them the same scalar polynomial. The identification summarized in Equation (1) is exact and basis-independent, but deliberately local to the attacker-imposed restriction (Theorem 4.1).

1.1 Our results

An exact symmetry quotient and public reduction. After the common-column restriction, every symmetric public base form factors through $\text{Sym}^2(A)$. This loss occurs before the outer public mixing, so later linear combinations cannot restore the missing forms. Public row reduction then separates the surviving quadratics from the affine conditions and target checks. For every published parameter shape that passes the stated public rank checks, this gives an exact quadratic system on a

public affine space. Every affine restriction preserves all residual equations, and every recovered root is reconstructed and checked by the original verifier ([Theorems 4.1 and 4.6](#) and [proposition 4.3](#)).

Conditional PXL estimates for $\ell = 2$. For $\ell = 2$, we prove full affine coefficient support on every transverse slice. A uniform target and affine-coset trial then has an exact first moment, and the spectrum theorem gives a lower bound on the probability of an accepted root with full column rank. We apply the five components from Section 4.2 of PXL under its ordinary MQ assumptions and a separate instancewise extraction premise. For $\omega = 2.81$, the resulting unit-leading gate exponents are 128.833, 187.030, and 245.042 across Levels I, III, and V ([Theorem 8.1](#)). The NIST reference exponents are 143, 207, and 272 [[16](#)]. Auxiliary operations remain explicit and unpriced, so these figures are not complete operation bounds.

The stronger $\ell = 4$ quotient and two PXL routes. For $\ell = 4$, symmetry removes six of every sixteen ordered quadratic forms. After splitting the four Frobenius blocks, we prove that the fixed coefficient map to the four diagonal and six cross channels is onto exactly when all four maps to the vinegar quotients are injective ([Theorem 6.5](#)). The official graph complement gives an exact zero-offset realization of this homogeneous support criterion. For positive-rank offsets, we state exact public compatibility checks as H_{off} ; their frequency under the official distribution is unproved. Under H_{off} and the instancewise PXL route premise, the support-constrained estimates cross the Level III and V reference exponents but miss Level I. A separate premise H_{struct} asserts the required Hilbert series, Macaulay ranks, degree, and extraction behavior directly for the structured channel family. Under that premise, the PXL estimates cross the reference exponents for all six $\ell = 4$ shapes ([Theorems 6.5 and 8.3](#)).

Design implications. We derive necessary output-dimension conditions for redesigns that retain the common-column architecture. These conditions are not sufficient for security ([Theorems 10.2 and 10.4](#)).

1.2 How to read the result

The paper separates the public reduction, the existence of acceptable roots, and the behavior of the algebraic solver. These statements have different logical status. The following table records the distinction used throughout the paper.

Table 1: Logical status of the attack. An exact statement is a deterministic claim about the supplied public key after its stated public checks pass.

Part	Status	Meaning
Column restriction and symmetry quotient	Exact	Public substitution followed by an algebraic identity.
Output split and affine elimination	Exact after public rank checks	Invertible row reduction preserves every verifier equation.
Affine coefficient support for $\ell = 2$	Exact on transverse slices	The coefficient map is onto. This does not imply uniform coefficients or semi-regularity.
Frequency of spectrum events	Distributional, or certified for a fixed key	The paper uses an idealized XOF-transcript model; a fixed key instead requires a spectrum certificate.
PXL regularity model	Conditional	The solving degree and matrix ranks use the ordinary MQ assumptions from PXL Section 4.2.
Instancewise PXL route	Conditional	The actual restricted system must follow the predicted ranks, and the added extraction step must recover a root when one exists.
$\ell = 4$ homogeneous fixed-map criterion	Exact after splitting	Surjectivity holds exactly under four quotient-injectivity conditions, and the official graph complement realizes the zero-offset case.
$\ell = 4$ positive-rank offsets	Conditional, with exact public checks	H_{off} requires $\text{rank}(C _{U_{\text{vin}}}) = \lambda$ and $A \ker(C _{U_{\text{vin}}}) = U_{\text{vin}}$; its official frequency is unproved.
Structured $\ell = 4$ PXL route	Conditional	H_{struct} directly assumes the required Hilbert series, degree, Macaulay ranks, and extraction behavior. It does not assert ordinary coefficient support.
PXL arithmetic	Unit-leading estimate	The five published components are aggregated conservatively. Auxiliary work and storage traffic remain unpriced.
Candidate acceptance	Exact	Every candidate is reconstructed and checked by the original verifier.
Practical time and memory	Not established	No production solve is given, and the analysis does not bound wall-clock time or peak memory.

1.3 Attack overview

In the hidden UOV coordinates, **SNOVA** contains no product of two oil blocks. The signer first chooses the vinegar blocks, after which the signing equations are linear in the oil entries. A secret invertible change of variables hides this division, so the public verifier presents a structured quadratic map rather than the linear system used by the signer (Section 2).

The attack seeks a direct public preimage, not the hidden oil space. It starts from the common-column relation used in earlier forgery attacks. Inside every matrix-valued signature block, the attacker writes all columns as public affine functions of one stacked column. This selects a public subset of possible signatures; it is not an identity of the unrestricted verifier (Section 4).

After this restriction, every ordered pair of public matrix multipliers gives a scalar quadratic polynomial. Symmetry makes the polynomials for the two orders equal. One group of sixteen ordered pairs therefore contributes at most ten distinct quadratics when $\ell = 4$, while one group of four contributes at most three when $\ell = 2$. The equality holds before the outer public map combines these forms, so later linear combinations cannot restore the missing quadratic directions (Theorem 4.1).

The attacker next computes an invertible public change of verifier outputs. Some new coordinates contain the surviving quadratic forms. The remaining coordinates are affine conditions on the signature and, when necessary, consistency conditions on the target. Solving the affine equations gives a public affine space. Restricting this space to as many variables as surviving quadratics gives a square system, but this is only one possible restriction. Any affine slice preserves every residual equation (Proposition 4.3 and corollary 4.4).

For $\ell = 2$, the attacker samples a transverse affine coset and keeps all residual equations. The exact support theorem shows that the native coefficient map reaches every ordinary affine MQ coefficient tuple on this slice. The root theorem bounds the chance that one target and coset trial contains an accepted root with full column rank. The conditional solver then applies PXL to the complete overdetermined system (Sections 6.1 and 8).

For $\ell = 4$, the exact quotient leaves ten channels per base form. The split fixed-map theorem identifies the precise condition under which these homogeneous channels have full support, and the official graph complement realizes it exactly at zero offset. The positive-rank route uses the public compatibility premise H_{off} . Its support-constrained PXL rows cross Levels III and V but not Level I. The cheaper structured route states its Hilbert-series, degree, Macaulay-rank, and extraction premise directly and gives conditional rows below the reference exponents for all six $\ell = 4$ shapes (Sections 6.2 and 8.3).

Finally, every recovered root is converted back into a candidate signature, checked against the public format rule, and passed to the original verifier. A failed rank test, target check, solve, or final verification causes a restart. The final check makes the attack one-sided: it can fail to return a signature, but it never returns an invalid one.

2 The SNOVA signature scheme

This section defines the part of Version 2.3 used in the attack. Fix parameters (v, o, q, ℓ, r) and put $n = v + o$. A vector of private or public variables has the form

$$X = (X_1, \dots, X_n) \in \text{Mat}_{\ell \times r}(\mathbb{F}_q)^n.$$

In the private coordinates, $X_1, \dots, X_v$ are the vinegar blocks and $X_{v+1}, \dots, X_n$ are the oil blocks. The central map has o outputs in $\text{Mat}_{r \times \ell}(\mathbb{F}_q)$, so both its oil input and its output contain $or\ell$ field elements [3, 4].

The hidden central map. The UOV structure excludes every product of two oil variables. Write $[n] = \{1, \dots, n\}$. The allowed ordered pairs are

$$\Omega = [n]^2 \setminus \{v+1, \dots, n\}^2. \quad (2)$$

Let $S \in \text{Mat}_{\ell}(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial, and let

$$A = \mathbb{F}_q[S] \cong \mathbb{F}_{q^\ell}.$$

Every element of A is a symmetric matrix, and all elements of A commute. Version 2.3 uses

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad N_\alpha = r\ell + r$$

base forms and outer summands. For each $h \in [m_1]$, the secret base form is an n by n block matrix $[F_h] = (F_{h,jk})$, where $F_{h,jk} \in \text{Mat}_{\ell}(\mathbb{F}_q)$ and $F_{h,jk} = 0$ whenever $(j, k) \notin \Omega$.

We write $L_{i,\alpha}$ and $R_{i,\alpha}$ for the matrices called $A_{i,\alpha}$ and $B_{i,\alpha}$ in the specification. Their dimensions are r by r and r by ℓ , respectively. The other public factors $Q_{i,\alpha,1}, Q_{i,\alpha,2}$ are nonzero elements of A . Put $h(i, \alpha) = 1 + ((i - 1 + \alpha) \bmod m_1)$. The i th component of the hidden central map is

$$\tilde{F}_i(X) = \sum_{\alpha=0}^{N_\alpha-1} L_{i,\alpha} \left(\sum_{(j,k) \in \Omega} X_j^\top Q_{i,\alpha,1} F_{h(i,\alpha),jk} Q_{i,\alpha,2} X_k \right) R_{i,\alpha}, \quad i \in [o]. \quad (3)$$

The inner sum is an r by r matrix, and the full output is an r by ℓ matrix. Since Equation (3) has no product of two oil blocks, fixing the vinegar blocks makes it linear in every entry of the oil blocks.

The secret change of variables and the public map. The secret matrix $T^{12} = (T_{ab})$ has v by o entries in A . It defines the invertible change of variables $X = T(U)$ by

$$\begin{aligned} X_a &= U_a - \sum_{b=1}^o T_{ab} U_{v+b}, \quad 1 \leq a \leq v, \\ X_{v+b} &= U_{v+b}, \quad 1 \leq b \leq o. \end{aligned} \quad (4)$$

Thus T hides the private vinegar and oil coordinates while preserving the matrix shape of each block. For every base form, key generation computes

$$[P_h] = [T]^\top [F_h] [T]. \quad (5)$$

The commutativity of A then gives the public quadratic map

$$\begin{aligned} \tilde{P}_i(U) &= \sum_{\alpha=0}^{N_\alpha-1} L_{i,\alpha} \left(\sum_{j,k=1}^n U_j^\top Q_{i,\alpha,1} P_{h(i,\alpha),jk} Q_{i,\alpha,2} U_k \right) R_{i,\alpha} \\ &= \tilde{F}_i(T(U)). \end{aligned} \quad (6)$$

For odd q , the public base forms are symmetric. In block notation, this means

$$P_{h,kj} = P_{h,jk}^\top. \quad (7)$$

Equivalently, expanding all block entries over $\mathbb{F}_q$ gives a symmetric $n\ell$ by $n\ell$ matrix P_h . This is the symmetry used by our attack.

The block equations also explain the public key compression. Partition the base forms according to the vinegar and oil coordinates:

$$[F_h] = \begin{bmatrix} F_h^{11} & F_h^{12} \\ F_h^{21} & 0 \end{bmatrix}, \quad [T] = \begin{bmatrix} I & -T^{12} \\ 0 & I \end{bmatrix}, \quad [P_h] = \begin{bmatrix} P_h^{11} & P_h^{12} \\ P_h^{21} & P_h^{22} \end{bmatrix}.$$

The public seed determines the outer matrices, the Q matrices, and $P_h^{11}, P_h^{12}, P_h^{21}$. The private seed determines T^{12} . Key generation reconstructs the hidden blocks and the remaining public block by

$$\begin{aligned} F_h^{11} &= P_h^{11}, & F_h^{12} &= P_h^{12} + P_h^{11} T^{12}, \\ F_h^{21} &= P_h^{21} + (T^{12})^\top P_h^{11}, & P_h^{22} &= -(T^{12})^\top P_h^{12} - P_h^{21} T^{12}. \end{aligned} \quad (8)$$

Thus the compressed public key is the public seed together with the blocks P_h^{22} for $h \in [m_1]$. The secret key contains the seeds needed to regenerate T and the hidden forms [4].

Signing and verification. For a message M and a 16-byte salt, the target is

$$Y = H(s_{\text{public}} \parallel \text{SHAKE256}(M, 64) \parallel \text{salt}) \in \text{Mat}_{r \times \ell}(\mathbb{F}_q)^o. \quad (9)$$

The signer derives the vinegar blocks from the private seed, M , the salt, and an attempt counter. Write $V = (X_1, \dots, X_v)$ for the chosen vinegar blocks and let z list the $or\ell$ entries of the oil blocks. By Equation (2), the signing equation is a square linear system

$$G_V z = \text{vec}(Y) - c_V, \quad (10)$$

where $c_V = \text{vec}(\tilde{F}(V, 0))$ and G_V is the coefficient matrix of the terms containing one vinegar block and one oil block. If G_V is singular, the signer chooses a new attempt. Otherwise it solves for the oil blocks, sets $U = T^{-1}(X)$, and retries if U violates the public format rule. The signature is $U \parallel \text{salt}$.

The verifier expands the public key, checks the same format rule, recomputes Y , and accepts exactly when

$$\tilde{P}(U) = Y. \quad (11)$$

Version 2.1 introduced the odd-prime fields and Equation (7). Version 2.3 retained that symmetry and allowed $r \neq \ell$, which gives rectangular signature blocks.

Column form of the public verifier. For each column $c \in \{0, \dots, r-1\}$, stack the c th columns of the public signature blocks as

$$u_c = (U_1[:, c]^T, \dots, U_n[:, c]^T)^T \in \mathbb{F}_q^{n\ell}. \quad (12)$$

Using the scalar expansion of $[P_h]$ and $\Sigma_\xi = I_n \otimes \xi$, each scalar entry inside Equation (6) is a linear combination of terms

$$u_c^T \Sigma_\xi P_h \Sigma_\eta u_d, \quad \xi, \eta \in A. \quad (13)$$

The outer matrices $L_{i,\alpha}$ and $R_{i,\alpha}$ combine these terms by a public linear map. Thus SNOVA's matrix structure affects the scalar verifier through the ordered left and right algebra multipliers in Equation (13). Section 4 imposes the attacker's column relation and determines which of these ordered terms remain distinct.

3 Prior attacks and mathematical background

3.1 SNOVA's attack and revision history

The first line of SNOVA cryptanalysis attacks the hidden UOV structure. Ikematsu and Akiyama analyze key recovery on the scalar expansion of the core map, and Li and Ding sharpen the interpretation as a UOV system with ℓv vinegar variables and ℓo oil variables over the base field [5, 6]. Nakamura, Tani, and Furue instead lift the same structure to a smaller UOV instance over $\mathbb{F}_{q^\ell}$ [10]. These attacks seek the hidden oil space or an equivalent signing key. They concern the original characteristic-two construction and do not use the symmetric public matrices introduced later.

Beullens exposes a different layer. He rewrites SNOVA as a structured MAYO-style whipping of a bilinear map, chooses an affine relation among the signature columns, and searches for rank-deficient combinations in the outer coefficient map [7]. The result is a forgery attack, including weak-key instances much cheaper than the generic case. The Round 2 changes vary the outer coefficient matrices and mitigate the largest rank drops, but they retain the affine-column attack surface.

Our attack starts from this public restriction and acts one layer earlier: symmetry identifies the restricted power-pair polynomials before the outer map sees them.

Cabarcas et al. and Furue et al. improve the algebraic solving of the systems arising in the earlier attacks [8, 9]. The former exploits stability under a commutative group action and adapts the result to reconciliation and direct forgery. The latter writes the extension-field transformation explicitly and applies the resulting multihomogeneous systems to intersection and rectangular-MinRank key recovery. Their estimate for the old Round 2 $(q, v, o, \ell) = (16, 29, 6, 5)$ set is below the claimed Level-V threshold, subject to their stated solving-degree heuristic. Both papers study the characteristic-two construction. Neither derives a loss of independent verifier equations from the later symmetric public matrices.

Ran’s wedge attack begins from the alternating polar forms of characteristic-two UOV and recovers the hidden oil space through exterior algebra [11]. His latest version also gives a different construction for odd characteristic, but states that SNOVA’s added structure prevents the analysis from carrying over directly. Bros et al. specialize the characteristic-two method to SNOVA and give conjectural estimates below the claimed security levels for every Round 2 parameter set [12]. The SNOVA team’s later analysis replaces the conjectural kernel formula with a generating function and finds six Round 2 sets below target [17].

Ding, Li, and Tao place these attacks in a common Macaulay-matrix framework and extend it to multihomogeneous systems in any characteristic [15]. Their formulation applies to SNOVA’s hidden oil-space problem, but their concrete SNOVA analysis is expressly limited to the even-characteristic Round 2 parameters. It does not analyze the symmetric public matrices or the nine $q = 19$ Version 2.3 shapes.

Jin et al. give the main prior attack that genuinely crosses the characteristic boundary [14]. Their p -reduced symmetric algebra recovers Ran’s exterior algebra at $p = 2$ and defines an XL key-recovery attack for arbitrary characteristic. Their concrete odd-characteristic application is to the twelve QR-UOV parameter sets, where the attack does not beat reconciliation on the recommended parameters. The method targets the hidden UOV oil space. It does not analyze the symmetric public-matrix encoding of odd-characteristic SNOVA, and it does not produce the verifier-preserving direct-preimage reduction used here.

The distinction between these attacks is summarized in Table 2. The exact new statement in this paper is the $\text{Sym}^2(A)$ factorization after the public common-column restriction and the resulting invertible change of verifier outputs. The direct-forgery cost estimates built on that reduction remain conditional on their stated root-count and solver hypotheses.

Table 2: Scope of the closest attacks. “Output” is what a successful attack recovers; it is not a claim that every cited estimate is practical.

Work	Targeted regime	Main mechanism	Output and relation to this paper
Ikematsu–Akiyama; Li–Ding; Nakamura– Tani–Furue	Original / early $q = 16$	Scalar or extension-field UOV interpretation	Oil-space or equivalent-key recovery; no odd- q public symmetrization.
Beullens; Cabarcas et al.; Furue et al.	Original and Round 2 $q = 16$	Affine-column rank drop, group action, and multihomogeneous solving	Direct forgery or key recovery; supplies the public restriction that this paper composes with symmetry.
Ran; Bros et al.; Tseng et al.; Ding–Li–Tao	Arbitrary- characteristic UOV frameworks; concrete SNOVA estimates for Round 2 $q = 16$	Exterior or symmetric forms, Macaulay matrices, and the wedge-map kernel	Oil-space recovery; no concrete analysis of the symmetric $q = 19$ SNOVA verifier.
Jin et al.	UOV families in arbitrary characteristic; concrete odd- q QR-UOV instances	p -reduced symmetric-algebra XL	Oil-space recovery; no analysis of symmetric odd- q SNOVA public matrices.
This paper	All nine $q = 19$ Version 2.3 shapes	Common-column restriction, exact $\text{Sym}^2(A)$ quotient, and public output split	Conditional direct preimages checked by the original verifier; no hidden-key recovery claim.

The complementary generic-MQ and partitioning literature includes Hashimoto’s underdefined optimizer, Just Guess, generalized partitioning, and recent F4 engineering [18–21]. Polynomial XL performs part of the Macaulay elimination over a polynomial ring before evaluating selected variables [22]. Its Section 4.2 estimates assume affine semi-regularity, projected semi-regularity, and predicted Macaulay ranks. Our solver model applies those standard premises to the exact quotient system and states the added instancewise extraction premise separately.

3.2 MQ systems, roots, and affine slices

An MQ system over $\mathbb{F}_q$ is a map $g : \mathbb{F}_q^N \rightarrow \mathbb{F}_q^K$ whose coordinates have total degree at most two, together with a target $z \in \mathbb{F}_q^K$. Linear and constant terms are allowed. A root is an x satisfying $g(x) = z$. Underdetermination makes roots more plentiful but does not make the system linear or automatically easy.

An affine space is a translate $a + V$ of a linear subspace. Restricting the search to an affine slice $a + L \subseteq a + V$ turns the problem into a system in $h = \dim L$ coordinates. Three events must be separated:

- (i) the target–slice pair contains a base-field root;
- (ii) it contains an accepted root at which a chosen Jacobian has full rank;
- (iii) the chosen algebraic solver recovers every such root within its stated work bound.

The first two are properties of the map and slice. The third is a solver statement. The later recovery analysis proves them separately.

4 From the public verifier to an affine quadratic system

This section starts from the column form of the public verifier in [Equations \(12\) and \(13\)](#) and gives the exact attack as one sequence of public operations. The attacker imposes the common-column relation. Symmetry then identifies restricted quadratic forms. An invertible change of verifier outputs separates the remaining quadratic equations from the affine ones. Once the public rank checks pass, the attacker may restrict the resulting affine space to any chosen dimension while keeping every residual equation. This section makes no claim that the resulting system has a root or that a solver can find one.

4.1 Parameters and public forms

Retain the notation of Section 2 and put

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell, \quad K = m_1 \binom{\ell+1}{2}. \quad (14)$$

The ceiling follows the pinned Version 2.3 reference implementation [\[4\]](#). The theorem below proves that K is the maximum number of distinct restricted quadratic forms left by symmetry.

For the public rows, $A = \mathbb{F}_q[S] \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. Write $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$ and use the symmetric scalar expansions $P_1, \dots, P_{m_1} \in \text{Mat}_{N_0}(\mathbb{F}_q)$ from [Equation \(5\)](#). For $\xi, \eta \in A$, define

$$B_i^{\xi, \eta}(x, y) = x^\top \Sigma_\xi P_i \Sigma_\eta y. \quad (15)$$

Setting $x = y$ gives the quadratic power-pair feature.

Table 3: Dimensions of the nine public $q = 19$ Version 2.3 parameter rows.

Level	(v, o, ℓ, r)	m_1	M	K	N_0
I	(28, 5, 4, 4)	5	80	50	132
I	(48, 16, 2, 2)	16	64	48	128
I	(28, 4, 4, 5)	5	80	50	128
III	(40, 7, 4, 4)	7	112	70	188
III	(72, 24, 2, 2)	24	96	72	192
III	(38, 5, 4, 5)	7	100	70	172
V	(50, 9, 4, 4)	9	144	90	236
V	(96, 32, 2, 2)	32	128	96	256
V	(52, 6, 4, 6)	9	144	90	232

4.2 The common-column restriction

Write the stacked signature columns as

$$U = [u_0 \mid \dots \mid u_{r-1}], \quad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses public multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$, and imposes

$$u_j = \Sigma_{R_j} u + v_j, \quad 0 \leq j < r. \quad (16)$$

This relation selects a public subset of possible signatures. It makes no assumption about the honest signer and does not change the public verifier.

Before substitution, the verifier feature vector has $N_{\text{raw}} = m_1 \ell^2 r^2$ labeled coordinates. Let $z_{\text{ord}}(u)$ list the remaining scalar quadratic forms indexed by ordered pairs of powers of S . Let E be the public outer map after the fixed permutation that reconciles the official flattening order with the block-transposed order used here. For a fixed relation R , let R_R expand $z_{\text{ord}}(u)$ into the raw feature vector, and let $F_{R,v}$ collect the coefficients of the terms that are linear in u . The substituted verifier is

$$E(R_R z_{\text{ord}}(u) + F_{R,v} u) + c_{R,v} = t. \quad (17)$$

Put

$$E_R = ER_R, \quad \Lambda_{R,v} = EF_{R,v}.$$

Products of two variable terms give $z_{\text{ord}}(u)$. Products of one variable term and one offset give the linear term, and products of two offsets give the constant.

4.3 How symmetry removes quadratic forms

For one base form in this restricted scalar map, the ordered label space is $A \otimes_{\mathbb{F}_q} A$. The following theorem identifies the pairs that always give the same polynomial.

Theorem 4.1 (Symmetric-square factorization). *Let $A = \mathbb{F}_q[S]$ have dimension d , and assume $S^\top = S$ and $P_i^\top = P_i$. For each i , the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\text{hom}}, \quad \xi \otimes \eta \longmapsto u^\top \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It factors through

$$\text{Sym}_{\mathbb{F}_q}^2(A) = (A \otimes_{\mathbb{F}_q} A) / \text{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently the direct sum over m_1 base forms has rank at most $m_1 \binom{d+1}{2}$. After any public linear map to M outputs, its rank is at most

$$\min \left\{ M, m_1 \binom{d+1}{2} \right\}. \quad (18)$$

If the minimal polynomial of S has degree ℓ , then $d = \ell$ and the ordered feature vector is obtained by duplicating the off-diagonal coordinates of an unordered feature vector.

Proof. Every element of A is symmetric. Since the displayed value is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = (u^\top \Sigma_\xi P_i \Sigma_\eta u)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. The symmetric square has dimension $\binom{d+1}{2}$, and a later linear map cannot enlarge its image. When powers of S form a basis, the coordinate statement is exactly the duplication of each off-diagonal unordered pair into its two ordered positions. $\square$

The theorem is an upper bound on the common-column scalar image: extra public dependencies can only reduce that image further. Every concrete route checks the exact public rank before solving. The outer map cannot repair the loss because, after the relation, it sees only the already-equal polynomial values. Before the common-column restriction, swapping (ξ, η) only transposes the matrix $U^\top \Sigma_\xi P_i \Sigma_\eta U$; it does not make the matrix entries equal. The theorem therefore concerns the restricted scalar verifier, not the unrestricted matrix-valued map. The factorization itself uses only transposition symmetry. Odd characteristic is needed later for the familiar decomposition into symmetric and alternating tensors and for the polar-form analysis. The submitted nonsymmetrized $q = 16$ construction is not covered.

4.4 Mapping the remaining forms to verifier outputs

Let $z_{\text{sym}}(u)$ contain one coordinate for each unordered pair $0 \leq a \leq b < \ell$ in every base form, and let D duplicate its off-diagonal coordinates. By [Theorem 4.1](#), $z_{\text{ord}}(u) = Dz_{\text{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M \times K}. \quad (19)$$

The columns of Q_R span every output direction reachable by the quotient quadratics.

4.5 Separating the quadratic and affine equations

Choose W_R whose rows form a basis of the left kernel of Q_R , and put

$$C_{R,v} = W_R \Lambda_{R,v}. \quad (20)$$

Multiplication by W_R removes every quadratic term and exposes the affine constraints. A complementary map G_R keeps the quadratic coordinates. The two maps together are an invertible change of verifier outputs.

Lemma 4.2 (Invertible output split). *Assume $\text{rank } Q_R = K$. Choose a public left inverse $G_R \in \mathbb{F}_q^{K \times M}$ satisfying $G_R Q_R = I_K$. Then*

$$\begin{pmatrix} G_R \\ W_R \end{pmatrix} \in \mathbb{F}_q^{M \times M}$$

is invertible. Writing $b = t - c_{R,v}$, the complete substituted verifier is equivalent to

$$C_{R,v}u = W_R b, \quad g_{R,v}(u) = G_R b, \quad g_{R,v}(u) = z_{\text{sym}}(u) + G_R \Lambda_{R,v} u. \quad (21)$$

The polar map of $g_{R,v}$ is exactly that of z_{sym} . If b is uniform in $\mathbb{F}_q^M$, the two right-hand sides in (21) are independent and uniform.

Proof. If a vector $y \in \mathbb{F}_q^M$ is killed by both G_R and W_R , then $W_R y = 0$ implies $y = Q_R a$ for some $a \in \mathbb{F}_q^K$. Applying G_R gives $a = G_R Q_R a = 0$, so $y = 0$. The stacked square matrix is therefore invertible. Applying its block rows to $Q_R z_{\text{sym}}(u) + \Lambda_{R,v} u = b$ gives the two displayed systems, and invertibility gives the converse. Affine terms have zero polar. Finally, an invertible linear transformation sends a uniform vector to a uniform vector on the product of the two coordinate spaces. $\square$

Proposition 4.3 (Exact affine reduction). *Let $\rho_Q = \text{rank } Q_R$, $\lambda = \text{rank } C_{R,v}$, and $\delta = M - \rho_Q - \lambda$. Then*

$$\text{rank}[Q_R \ \Lambda_{R,v}] = \rho_Q + \lambda.$$

After row reduction, exactly δ independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves ρ_Q equations of degree at most two in $N_0 - \lambda$ variables, with exactly the same solution set as the complete substituted verifier.

If $\rho_Q = K$ and $\lambda = M - K$, every target passes the affine consistency test and the residual system consists of K equations on an affine translate of $\ker C_{R,v}$.

Proof. The row space of W_R is $\ker Q_R^T$, so $\ker W_R = (\ker Q_R^T)^\perp = \text{im } Q_R$. Hence W_R realizes the quotient map $\mathbb{F}_q^M \rightarrow \mathbb{F}_q^M / \text{im } Q_R$. The image of the columns of $\Lambda_{R,v}$ in this quotient has dimension $\text{rank}(W_R \Lambda_{R,v}) = \lambda$, proving the first identity.

The projected affine system is $C_{R,v}u = W_R b$. Its codomain has dimension $M - \rho_Q$ and its image has dimension λ , leaving $\delta = (M - \rho_Q) - \lambda$ target consistency equations. Whenever they hold, parameterize the affine solution space by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ_Q rows. Every operation is invertible row reduction or exact affine elimination, so the solution set is unchanged. In the full-rank case, $C_{R,v}$ is surjective and $\delta = 0$. $\square$

4.6 Restricting the remaining equations to an affine slice

The preceding proposition preserves the full solution set inside the common-column relation. The attack may now choose a smaller affine space. This second restriction preserves every verifier equation, but it need not contain a solution. Root existence is handled separately in [Section 7](#).

Corollary 4.4 (Exact affine restriction). *Assume $\text{rank } Q_R = K$ and $\text{rank } C_{R,v} = M - K$. For every target t , choose one solution $u_\star$ to $C_{R,v}u = W_R(t - c_{R,v})$, an h -dimensional subspace $L \subseteq \ker C_{R,v}$, and an injection $T : \mathbb{F}_q^h \rightarrow L$. On the affine space $u_\star + L$, the complete substituted verifier is equivalent to*

$$g_{R,v}(u_\star + Tx) = G_R(t - c_{R,v}), \quad x \in \mathbb{F}_q^h. \quad (22)$$

This is a system of K affine quadratic equations in h variables. Every root reconstructs, through (16), a candidate that satisfies all equations in the original verifier.

Proof. The rank assumption makes $C_{R,v}$ surjective, so $u_\star$ exists. The affine equations in (21) hold throughout $u_\star + L$. The invertible output change in [Lemma 4.2](#) leaves exactly the K residual equations in (22). That lemma also proves the claimed equivalence. $\square$

Taking $h = K$ gives the square restriction used by the earlier solver analysis.

Corollary 4.5 (Exact square restriction). *Assume*

$$\text{rank } Q_R = K, \quad \text{rank } C_{R,v} = M - K, \quad N_0 \geq M.$$

Then, for every target t , choose one solution $u_\star$ to $C_{R,v}u = W_R(t - c_{R,v})$, a K -dimensional subspace $L \subseteq \ker C_{R,v}$, and an isomorphism $T : \mathbb{F}_q^K \rightarrow L$. On the affine space $u_\star + L$, the complete substituted verifier is equivalent to the square system

$$g_{R,v}(u_\star + Tx) = G_R(t - c_{R,v}), \quad x \in \mathbb{F}_q^K. \quad (23)$$

Every root reconstructs, through (16), a candidate that satisfies the original verifier equations. Conversely, every verifier preimage that satisfies both public restrictions gives a root of (23). The corollary does not assert that a root exists or that it passes the signature-format rejection rule.

Proof. The rank assumption on $C_{R,v}$ makes it surjective, so $u_\star$ exists for every target. Its kernel has dimension

$$N_0 - (M - K) = N_0 - M + K \geq K,$$

so it contains a K -dimensional subspace L . The affine equations in (21) hold throughout $u_\star + L$. The invertible output change in [Lemma 4.2](#) therefore leaves exactly the K quadratic equations in (23). That lemma also gives both directions of the claimed equivalence. $\square$

All nine published parameter shapes satisfy $N_0 \geq M$ by [Table 3](#). Thus, for these shapes, the two displayed rank checks are enough to construct the exact square system. They are not enough to show that this system contains an acceptable root.

Theorem 4.6 (Exact quotient for all published shapes). *For every published $q = 19$ Version 2.3 shape in Table 3, the common-column restriction factors each public base form through $\text{Sym}^2(A)$. After the stated public rank checks, the complete verifier is equivalent to $K = m_1 \binom{\ell+1}{2}$ affine quadratic equations on a public affine space. For $\ell = 4$, this changes sixteen ordered forms to ten per base form. For $\ell = 2$, it changes four ordered forms to three. Every further affine slice preserves all K residual equations, and every returned candidate is checked by the original verifier.*

Proof. Apply Theorem 4.1, proposition 4.3, and corollary 4.4 to each row of Table 3. $\square$

Operational meaning. The attacker computes every matrix in this section from the supplied public key. The attacker may test several public relations and offsets before solving, but may not replace an unfavorable key. This paper does not prove how often that search succeeds or measure its cost separately. Those costs belong to C_{aux} for the current PXL model. If no tested relation passes the two rank checks, the conditional recovery result does not apply to that key. Once the checks pass, the steps in this section are exact public computations and no verifier equation is dropped.

4.7 A Level-I dimension walkthrough

For $(v, o, \ell, r) = (28, 5, 4, 4)$, we have $n = 33$, $N_0 = 132$, $M = 80$, $m_1 = 5$, and $K = 50$. The common-column relation leaves 80 ordered quadratic labels, but the symmetry quotient leaves only 50 distinct quadratics. In the full-rank affine case, the other 30 output directions become linear equations and the full affine residual domain has dimension $132 - 30 = 102$. The attacker can therefore choose a 50-dimensional affine space and obtain 50 quadratic equations in 50 variables. This is the exact reduction. Whether the chosen space contains an acceptable root, and how much work is needed to find one, are the subjects of the conditional analysis later in the paper.

5 Slice preparation and target sampling

The exact reduction permits any suitable affine slice inside $\ker C_{R,v}$. Recovery needs a slice chosen without reading its fresh coefficients, signatures that pass the format rule, and uniform targets. This section provides those inputs; root existence is addressed in Section 7.

5.1 An unconditional stable kernel

Generic offsets can supply the missing affine output directions, but their linear terms may destroy the block grading used by multihomogeneous solvers. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers $\Gamma_0, \dots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \quad (24)$$

Define the offset row space

$$U_w = \text{span}_{\mathbb{F}_q} \{w^\top \Sigma_\xi P_i \Sigma_\eta : 1 \leq i \leq m_1, \xi, \eta \in A\}. \quad (25)$$

For a basis $1, S, \dots, S^{d-1}$ of A , form the complete right- A orbit

$$O_{R,v}^F = \begin{pmatrix} F_{R,v} \\ F_{R,v} \Sigma_S \\ \vdots \\ F_{R,v} \Sigma_{S^{d-1}} \end{pmatrix}, \quad \rho_F = \text{rank } O_{R,v}^F. \quad (26)$$

Theorem 5.1 (Stable-lift kernel). *Assume $S^\top = S$, $P_i^\top = P_i$, and $A \cong \mathbb{F}_{q^d}$. Under (24):*

- (i) U_w is closed under right multiplication by A and $\dim_{\mathbb{F}_q} U_w \leq m_1 d^2$;
- (ii) every row of $F_{R,v}$ and $O_{R,v}^F$ lies in U_w ;
- (iii) the row space of $O_{R,v}^F$ is an A -subspace, so $d \mid \rho_F$;
- (iv)

$$H_{R,v}^F = \ker O_{R,v}^F$$

is A -linear, satisfies $F_{R,v} H_{R,v}^F = 0$, and has

$$\dim_A H_{R,v}^F = n - \rho_F/d \geq n - m_1 d. \quad (27)$$

No generic orbit-rank equality is required. An unexpected dependency only enlarges the stable space.

Right A -stability puts the orbit rows in an A -linear row space, and rank-nullity gives (27). The complete proof is in [Section B](#).

Since $C_{R,v} = W_R E F_{R,v}$, the stable kernel lies inside $\ker C_{R,v}$. Translation along a subspace $L \subseteq H_{R,v}^F$ changes the quotient quadratics but leaves every offset-linear feature constant.

5.2 Reserved-line decoupling and the structural preflight

Let $e_1, \dots, e_n$ be the native A -coordinate basis in which the key-dependent symmetric public matrices are expanded. Fix a vinegar index j_0 and use the coordinate-aligned decomposition

$$\begin{aligned} W &= A e_{j_0}, & V' &= \bigoplus_{\substack{j \text{ vinegar} \\ j \neq j_0}} A e_j, & O_{\text{pub}} &= \bigoplus_{j \text{ oil}} A e_j, \\ V_{\text{pub}} &= W \oplus V' \oplus O_{\text{pub}}, & \dim_A W &= 1, & \dim_A V' &= v - 1, & \dim_A O_{\text{pub}} &= o. \end{aligned} \quad (28)$$

The actual SHAKE stream is deterministic once the public seed is fixed. For probability statements only, the *native-coordinate idealized XOF-transcript model* replaces the relevant bytes by independent uniform bytes and reduces the native upper-triangular entries of the key-dependent P_i modulo 19. This is not a claim about a fixed SHAKE-generated key. The model conditions on E , R , and the full vector $\Gamma = (\Gamma_0, \dots, \Gamma_{r-1})$, and invokes independence before any A -linear basis change. A reduced byte has maximum atom

$$\rho_{\text{byte}} = \frac{14}{256},$$

because nine residues have fourteen preimages and ten have thirteen.

Lemma 5.2 (Coordinate-aligned reserved-line decoupling). *Fix E, R, Γ , choose $0 \neq w \in W$, and use the structured offsets (24). Then Q_R depends only on E and R , while $C_{R,v}$, $O_{R,v}^F$, $H_{R,v}^F$, and every deterministically tie-broken slice selected from them depend only on native entries having at least one index in W . In the native-coordinate idealized XOF-transcript model they are independent of the native $V' \times V'$ entries.*

Every offset contains one coordinate from W , so the construction never reads a native $V' \times V'$ entry. The complete proof is in [Section B](#).

Lemma 5.3 (Maximum-atom pivot bound). *Let $X_1, \dots, X_N$ be independent finite-field variables, each with maximum atom at most ρ_{byte} . If a matrix B has rank t , then for every c ,*

$$\Pr[BX = c] \leq \rho_{\text{byte}}^t.$$

Conditioning on the nonpivot variables leaves one allowed value for each pivot, so the probabilities multiply. The complete proof is in [Section B](#).

Put $\mathcal{Q}_R = \mathbb{F}_q^M / \text{im } Q_R$. When $\text{rank } Q_R = K$, $\dim_{\mathbb{F}_q} \mathcal{Q}_R = M - K$, and $C_{R,v}$ is a coordinate matrix for the induced linear map $u \mapsto W_R \Lambda_{R,v} u$ to $\mathcal{Q}_R$. Let $\mathbf{X}_{W,V'}$ be the vector of native independent public-matrix entries in the $W \times V'$ blocks. For $0 \neq \lambda \in \mathcal{Q}_R^\vee$, define B_λ by

$$\text{vec}((\lambda \circ C_{R,v})|_{V'}) = B_\lambda \mathbf{X}_{W,V'} + b_\lambda.$$

Here B_λ and b_λ are fixed once E, R, Γ, λ and the native layout are fixed; they do not depend on the sampled entries in $\mathbf{X}_{W,V'}$. Scaling λ scales B_λ , so its rank depends only on $[\lambda]$. Put

$$t_{\text{src}} = \min_{0 \neq \lambda \in \mathcal{Q}_R^\vee} \text{rank } B_\lambda.$$

For an A -line $\mathfrak{l} \subseteq W \oplus O_{\text{pub}}$, choose $0 \neq y_{\mathfrak{l}} \in \mathfrak{l}$ and write

$$O_{R,v}^F y_{\mathfrak{l}} = D_{\mathfrak{l}} \mathbf{X}_{\text{nat}} + e_{\mathfrak{l}}$$

in the complete vector $\mathbf{X}_{\text{nat}}$ of native independent public-matrix entries. The coefficient data $D_{\mathfrak{l}}, e_{\mathfrak{l}}$ are deterministic before those entries are sampled, and the rank of $D_{\mathfrak{l}}$ is independent of the chosen generator. Define

$$t_W = \text{rank } D_W, \quad t_{\text{oth}} = \min_{\substack{\mathfrak{l} \subseteq W \oplus O_{\text{pub}} \\ \dim_A \mathfrak{l} = 1, \mathfrak{l} \neq W}} \text{rank } D_{\mathfrak{l}}.$$

These are exact symbolic coefficient-map ranks determined by the fixed relation, the complete offset pattern, and the native coordinate layout; they do not depend on the sampled values of the key coefficients. Define

$$\delta_{\text{src}} = \frac{q^{M-K} - 1}{q - 1} \rho_{\text{byte}}^{d(v-1)}, \quad (29)$$

$$\delta_{\text{proj}} = \rho_{\text{byte}}^{m_1 \binom{d+1}{2}} + \left(\frac{|A|^{1+o} - 1}{|A| - 1} - 1 \right) \rho_{\text{byte}}^{m_1 d^2}, \quad |A| = q^d. \quad (30)$$

Theorem 5.4 (Conditional coefficient-preflight density). *Fix the deterministic outer map E , a relation R , and the complete vector Γ . Assume the exact deterministic checks*

$$\text{rank } Q_R = K, \quad t_{\text{src}} \geq d(v-1), \quad t_W \geq m_1 \binom{d+1}{2}, \quad t_{\text{oth}} \geq m_1 d^2$$

pass. In the native-coordinate idealized XOF-transcript model,

$$\Pr[\text{rank } C_{R,v} < M - K] \leq \delta_{\text{src}},$$

and

$$\Pr[H_{R,v}^F \cap (W \oplus O_{\text{pub}}) \neq 0] \leq \delta_{\text{proj}}.$$

Consequently full affine rank and injectivity of $H_{R,v}^F \rightarrow V'$ hold jointly with probability at least $1 - \delta_{\text{src}} - \delta_{\text{proj}}$. For the six official $\ell = 4$ dimension tuples, direct substitution gives $\delta_{\text{src}} + \delta_{\text{proj}} < 2^{-209}$. The coefficient-map ranks are fixed-layout symbolic checks; the realized quotient, affine, and intersection ranks are exact public checks on a fixed key.

The two bounds apply [Lemma 5.3](#) to projective output directions and to A -lines in $W \oplus O_{\text{pub}}$, then take union bounds. The complete proof is in [Section B](#).

The concrete check records all r entries of Γ and runs the displayed tests. This release does not assert that every fixed key passes them. This reserved-line preflight is separate from H_{off} and H_{struct} and does not establish either PXL route premise.

5.3 The guaranteed fresh ordinary square

Let

$$D_0 = V' \cap \ker C_{R,v}.$$

The intersection inequality gives

$$\dim_{\mathbb{F}_{19}} D_0 \geq 4(v-1) - (M-K). \quad (31)$$

For all six $\ell = 4$ rows this lower bound is at least K :

Parameters	$\dim D_0$ lower bound	K
$(28, 5, 4, 4), (28, 4, 4, 5)$	78	50
$(40, 7, 4, 4)$	114	70
$(38, 5, 4, 5)$	118	70
$(50, 9, 4, 4)$	142	90
$(52, 6, 4, 6)$	150	90

Consequently the attacker can choose, from cross data alone, an ordinary K -dimensional slice inside D_0 . Its internal $V' \times V'$ coefficients remain fresh for the spectrum theorem. This supplies an auxiliary square slice for every $\ell = 4$ row, including $(38, 5, 4, 5)$. The PXL tables use different dimensions and premises.

5.4 Passing the block-symmetry rejection rule

Version 2.3 rejects a square $\ell = 4$ signature if any block is symmetric and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [4]. Rectangular $\ell = 4$ blocks are not subject to the square-block rule.

For a square $\ell = 4$ relation $R = (R_0, R_1, R_2, R_3)$, define the local skew map

$$\mathcal{S}_R : \mathbb{F}_q^4 \longrightarrow \bigwedge^2 \mathbb{F}_q^4, \quad \mathcal{S}_R(x)_{a,c} = (R_c x)_a - (R_a x)_c.$$

For any offsets, one signature block is symmetric exactly when $\mathcal{S}_R(x)$ equals a fixed offset-dependent vector. The public relation $R_j = S^j$ for the official matrix has $\text{rank } \mathcal{S}_R = 4$. Thus each block has at most one rejected common-column value, and the accepted density in the full common-column space is at least

$$\alpha_{4,n} = (1 - 19^{-4})^n. \quad (32)$$

The accepted-root theorem later uses only this global density; it assumes no independence from the MQ system.

For $\ell = 2$, the zero-offset relation used in the main theorem gives a block

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

which is symmetric exactly when $y = 0$. The exact accepted density is derived in (36). A deterministic fixed-skew alternative is recorded in Section B; it is not used by the main theorem.

5.5 Exactly uniform official targets

The probability theorems require a uniform target over $\mathbb{F}_{19}^M$, but the official binary-to-field conversion has a small modulo bias. Model the target hash/XOF chunks as independent uniform bit strings, separately from the key expansion above. The pinned conversion uses each full eight-byte chunk for fifteen base-19 digits and a final partial chunk for the remainder. If a uniform b -bit chunk X supplies a digits, retain the message-salt input only when

$$X < \left\lfloor \frac{2^b}{19^a} \right\rfloor 19^a. \quad (33)$$

Conditioned on this event, $X \bmod 19^a$ is uniform. Applying the test to each chunk makes the retained official target exactly uniform, and an invertible output change preserves uniformity. Exact evaluation over the nine public Version 2.3 lengths gives a minimum acceptance probability $0.1524622985\dots$, and hence at least 0.152462 . When target consistency requires more than the 128-bit salt space, the existential chosen-message attacker varies a message prefix as well as the salt. Target generation occurs before nonlinear solving and adds to, rather than multiplies, the solve cost.

The exact $\ell = 2$ zero-offset split. For the three official $\ell = r = 2$ rows put

$$A = \mathbb{F}_{19^2}, \quad m = m_1 = o, \quad M = 4m, \quad K = 3m,$$

and use the public relation

$$u_0 = u, \quad u_1 = 0. \quad (34)$$

Proposition 5.5 (Exact zero-offset target split). *Let Q_R be the quotient-output matrix for (34). The route applies to a fixed key only after the exact public test rank $Q_R = K$. After a passing test, choose a left inverse G_R and a full-row-rank matrix W_R spanning the left kernel of Q_R . The complete substituted verifier is equivalent to*

$$W_R t = 0, \quad z_{\text{sym}}(u) = G_R t. \quad (35)$$

For a uniform official target, consistency has probability $19^{-(M-K)} = 19^{-m}$. Conditioned on consistency, the residual target is uniform in $\mathbb{F}_{19}^K$, so the attacker can filter message and salt inputs before solving. Under the stipulated 2^{32} target-charge convention in (75), this contributes $2^{32}19^m$ binary-gate units to the model numerator. This is an accounting convention, not a measured target-generation bound.

Proof. Zero offsets remove the affine and constant terms, leaving $Q_R z_{\text{sym}}(u) = t$. The invertible output change with block rows G_R, W_R gives (35). It preserves uniformity and separates the residual target from the consistency coordinates. Since target filtering precedes the nonlinear solve, its work adds to the solve ledger. $\square$

In one block, (34) has matrix $\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix}$ and is symmetric exactly when $y = 0$. The $n = v + o$ block conditions use disjoint coordinates. Thus the exact density accepted by the Version 2.3 rule is

$$\alpha_n = 19^{-n} \sum_{j=0}^{\lfloor n/4 \rfloor} \binom{n}{j} 18^{n-j}. \quad (36)$$

6 Affine coefficient support for the PXL routes

The PXL analysis keeps every residual equation and permits an overdetermined affine slice. We first prove full affine coefficient support for $\ell = 2$. We then give the exact fixed-map criterion that the $\ell = 4$ slice family must satisfy.

6.1 Transverse slices for $\ell = 2$

For the three official $\ell = r = 2$ shapes, put $A = \mathbb{F}_{19^2}$ and identify A^{v+o} with $\mathbb{F}_{19}^{2(v+o)}$. Write $A^{v+o} = V_{\text{pub}} \oplus O_{\text{pub}}$, where $V_{\text{pub}} \cong A^v$, and let π_v be the projection to V_{pub} . Under the zero-offset relation (34), the public rank test in Proposition 5.5 leaves $K = 3m$ residual equations and an exact target-consistency test. For Level I, this is a map

$$g : A^{64} \longrightarrow \mathbb{F}_{19}^{48}. \quad (37)$$

An injection $T : \mathbb{F}_{19}^h \rightarrow A^{v+o}$ defines the direction $L = \text{im } T$. For an $\mathbb{F}_{19}$ basis $\ell_1, \dots, \ell_h$ of L , put

$$L_A = \text{span}_A\{\pi_v(\ell_1), \dots, \pi_v(\ell_h)\} \subseteq V_{\text{pub}}. \quad (38)$$

We call L *A-independent* if $\dim_A L_A = h$. We call an affine coset $a + L$ *transverse* if

$$\pi_v(a) \notin L_A. \quad (39)$$

Both conditions are stronger than base-field injectivity alone.

Lemma 6.1 (Uniform coset sampling). *Let U be an $\mathbb{F}_q$ space and let $L \subseteq U$ have dimension h . If a is uniform in U , then $a + L$ is uniform in U/L . For every $x \in U$,*

$$\Pr[x \in a + L] = q^{h - \dim U}. \quad (40)$$

If z is independently uniform in $\mathbb{F}_q^K$, then

$$\Pr[x \in a + L \text{ and } g(x) = z] = q^{h - \dim U - K}. \quad (41)$$

If $U = A^{v+o}$ and L is A-independent, then

$$\Pr[\pi_v(a) \notin L_A] = 1 - |A|^{h-v}. \quad (42)$$

Proof. Every coset has q^h representatives, which proves the first claim and (40). Independence of z contributes q^{-K} in (41). The projection $\pi_v(a)$ is uniform in A^v , while L_A has $|A|^h$ elements. This proves (42). $\square$

Proposition 6.2 (Exact transverse affine restriction). *Assume the public rank check in Proposition 5.5. For every consistent target and every affine coset $a + \text{im } T$, the complete substituted verifier is equivalent on that coset to*

$$f_{a,z}(x) = g(a + Tx) - z = 0, \quad x \in \mathbb{F}_{19}^h. \quad (43)$$

This is a system of K affine quadratic equations in h variables. Every accepted root reconstructs a candidate that satisfies every equation in the original verifier.

Proof. The target split is an invertible public change of outputs. The map $x \mapsto a + Tx$ is an affine injection, so substitution preserves the solution set in the coset. The reconstruction uses (16), followed by the format rule and the original verifier. $\square$

Lemma 6.3 (Mixed block support). *Let $C : A \times A \rightarrow \mathbb{F}_{19}$ be bilinear and put $C^{\text{sym}}(\xi, \zeta) = C(\xi, \zeta) + C(\zeta, \xi)$. The map $C \mapsto C^{\text{sym}}$ is onto the symmetric bilinear forms on the two-dimensional $\mathbb{F}_{19}$ space A . Its kernel is the one-dimensional space of alternating forms.*

Proof. Since the characteristic is odd, a symmetric form S has preimage $S/2$. The kernel consists exactly of alternating forms on A . $\square$

Proposition 6.4 ($\ell = 2$ affine coefficient support). *Let $h \leq v - 1$, let L be A -independent, and let $a + L$ be transverse. For one fresh UOV base form, the restriction map from its fresh native symmetric coefficients to the three descended $\ell = 2$ channel polynomials on $a + L$ is onto*

$$(\mathbb{F}_{19}[x_1, \dots, x_h]_{\leq 2})^3. \quad (44)$$

Across the m independent base forms, the coefficient map is onto the space of $K = 3m$ ordinary affine quadratic equations in h variables. In particular, $h = v - 2$ has transversality failure probability $|A|^{-2} = 19^{-4}$.

Proof. Put $b = \pi_v(a)$ and $e_i = \pi_v(\ell_i)$. Transversality and A -independence give the direct sum

$$Ab \oplus Ae_1 \oplus \dots \oplus Ae_h \subseteq V_{\text{pub}}.$$

The diagonal block on Ae_i supplies the three channel coefficients of x_i^2 . The block between Ae_i and Ae_j supplies all three coefficients of $x_i x_j$ by Lemma 6.3. The blocks between Ab and Ae_i supply the linear coefficients, and the diagonal block on Ab supplies the constants. These blocks are independent, so every triple of affine quadratics has a native preimage.

The block map is defined over $\mathbb{F}_{19}$. Its surjectivity after scalar extension therefore descends by faithful flatness. Taking the direct sum over the independent base forms gives all K equations. The final probability is (42) with $h = v - 2$. $\square$

This proposition proves support, not distribution. Uniform output follows from a uniform native source, but the official byte-reduction source is not uniform on $\mathbb{F}_{19}$. The proposition does not prove semi-regularity or any Macaulay rank prediction.

6.2 Fixed-map support for $\ell = 4$

Work over a splitting field Ω of odd characteristic. Let the four Frobenius blocks be V_a , with planted oil spaces $O_a \subseteq V_a$, and fix maps $T_a : X \rightarrow V_a$ for $0 \leq a < 4$. Put $\bar{T}_a : X \rightarrow V_a/O_a$. The allowed diagonal coefficient block consists of symmetric forms on V_a that vanish on $O_a \times O_a$. The allowed cross block for $a < b$ consists of bilinear forms on $V_a \times V_b$ that vanish on $O_a \times O_b$. Pullback gives a fixed linear map Φ_T to four diagonal and six cross quadratic forms on X .

Theorem 6.5 (Fixed-map channel surjectivity). *The map*

$$\Phi_T : \mathcal{U}_O \longrightarrow \text{Sym}^2(X^*)^{10}$$

is onto if and only if every $\bar{T}_a$ is injective.

Proof. Suppose every $\bar{T}_a$ is injective. Choose complements $V_a = O_a \oplus C_a$. The projection of $T_a(X)$ to C_a is an isomorphic copy of X . Any symmetric form on X can therefore be transported to that copy, extended to C_a , and set to zero on blocks involving O_a . This realizes every diagonal channel.

For $a < b$, the two projected copies of X are injective. Given $S \in \text{Sym}^2(X^*)$, prescribe the cross bilinear form to be $S/2$ on these copies, extend it to $C_a \times C_b$, and set blocks involving the oil spaces

to zero. Its symmetric pullback is S . The ten source blocks are independent, so their direct sum is onto.

Conversely, if $0 \neq x \in \ker \bar{T}_a$, then $T_a x \in O_a$. Every allowed diagonal form vanishes at $(T_a x, T_a x)$, so that diagonal channel cannot realize a quadratic form that is nonzero at x . $\square$

For both Level I $\ell = 4$ shapes, $\dim(V_a/O_a) = 28$. A fixed PXL main space of dimension 31 therefore cannot satisfy this criterion. Full support through this theorem requires a main space of dimension at most 28, together with a realizable slice for which all four quotient maps are injective. If the spaces, maps, and coefficient conditions arise by scalar extension from $\mathbb{F}_{19}$, surjectivity descends by faithful flatness. An eigenblock decomposition over Ω alone does not supply that descent or a base-field realization.

The official change of variables supplies the required realization. Write the public coordinate space as

$$E = U_{\text{vin}} \oplus O_{\text{pub}}, \quad U_{\text{vin}} = A^v \oplus 0, \quad O_{\text{pub}} = 0 \oplus A^o,$$

and put $R = T^{12}$. The hidden oil space in public coordinates is the graph

$$G = \{(Rz, z) : z \in A^o\}. \quad (45)$$

Lemma 6.6 (Official graph complement). *The public vinegar space is a complement to the hidden oil graph:*

$$E = U_{\text{vin}} \oplus G. \quad (46)$$

After scalar extension to a splitting field, let E_a, U_a, G_a denote the four Frobenius components. For every a , the quotient map

$$U_a \longrightarrow E_a/G_a \quad (47)$$

is an isomorphism.

Proof. Every $(x, z) \in E$ has the unique decomposition

$$(x, z) = (x - Rz, 0) + (Rz, z).$$

This proves (46). Scalar extension and projection to each primitive idempotent preserve direct sums. Thus $E_a = U_a \oplus G_a$, which is equivalent to (47). $\square$

Choose $d \leq v$ public vinegar coordinate vectors that are A -independent, and let X be their $\mathbb{F}_{19}$ span. Each split projection of X maps injectively to E_a/G_a by Lemma 6.6. Therefore Theorem 6.5 applies simultaneously to all four Frobenius components. This proves that the official realization family meets the fixed-map good locus whenever the PXL main dimension is at most v .

For a full PXL slice, choose a direct sum

$$L = X \oplus Z, \quad \dim X = d, \quad \dim Z = k, \quad (48)$$

and a public coordinate map $\Gamma : L \rightarrow \mathbb{F}_{19}^k$ that kills X and is an isomorphism on Z . Fixing the Γ coordinates leaves an affine translate of X . Its leading quadratic forms are the pullbacks analyzed by Theorem 6.5. The constrained ledger below chooses $d = h - k = v$ for every row.

Zero offsets put $C_{R,0} = 0$, so every such L lies in the residual affine kernel. To avoid the resulting target-consistency cost, the supported route uses the following separate offset premise.

Definition 6.7 (Oil-supported offset premise H_{off}). *For the selected public relation, there are publicly specified oil-supported offsets for which the exact checks*

$$\text{rank}(C_{R,v}|_{U_{\text{vin}}}) = \lambda, \quad A \ker(C_{R,v}|_{U_{\text{vin}}}) = U_{\text{vin}} \quad (49)$$

hold, and the construction in (48) can be performed inside $\ker C_{R,v}$, where $\lambda = M - K$ is the full-rank value used by the ledger. Here AW denotes the A span of an $\mathbb{F}_{19}$ subspace W . The matrices and both displayed conditions are exact public checks. The premise makes no claim about how often official keys or sampled offsets pass them.

Under H_{off} , the graph complement proves homogeneous channel support on the PXL main variables. It does not prove full affine coefficient support. Semi-regularity, Macaulay ranks, and extraction remain conditional on H_{route} . The frequency of the offset premise under the official key distribution is unproved.

7 Root existence on an affine slice

The preceding sections produce affine MQ systems but do not show that a chosen slice has a root with the required format and a full-column-rank Jacobian. This section derives that probability from an exact collision identity and bounds the required derivative ranks for the slices used later.

7.1 Polar forms, rank spectra, and collisions

For a scalar quadratic polynomial f , define the polar form

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic this is bilinear. For a vector-valued map g , a nonzero $b \in \mathbb{F}_q^K$ selects the scalar quadratic $b \cdot g$. Its polar rank measures how many independent directions the quadratic part couples.

For a fixed slice direction y , the vector-valued derivative

$$T_y(x) = g(x + y) - g(x) - g(y) + g(0) \quad (50)$$

is linear in x . The value $q^{-\text{rank } T_y}$ is exactly the fraction of its codomain-dual directions that annihilate T_y , after the appropriate normalization. Summing these quantities over nonzero y controls the second factorial moment of the number of roots in a random target-coset pair. The same spectrum controls the density of points where the restricted Jacobian loses rank.

7.2 The exact first and second moments

Theorem 7.1 (Root-collision identity). *Let U be an $\mathbb{F}_q$ -space, let $C : U \rightarrow \mathbb{F}_q^c$ be surjective, let $L \subseteq \ker C$ have dimension h , and let $g : U \rightarrow \mathbb{F}_q^K$ have degree at most two. Sample independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, then choose a uniform affine coset $\mathcal{A}$ of L inside $C^{-1}(d)$. Put*

$$Z = |\{x \in \mathcal{A} : g(x) = z\}|, \quad \mu = q^{h-K}.$$

For $0 \neq y \in L$, define

$$T_y(x) = g(x + y) - g(x) - g(y) + g(0)$$

and let $\kappa(y)$ be one if $-(g(y) - g(0)) \in \text{im } T_y$ and zero otherwise. Then

$$\mathbb{E}Z = \mu, \quad (51)$$

$$\mathbb{E}[Z(Z-1)] = \mu \sum_{0 \neq y \in L} \kappa(y) q^{-\text{rank } T_y} \leq \mu \sum_{0 \neq y \in L} q^{-\text{rank } T_y}. \quad (52)$$

The domain of T_y is the full space U because the experiment averages over the affine right-hand side as well as the coset.

Proof. Sampling d and then a uniform L -coset in $C^{-1}(d)$ is exactly uniform over all cosets of L in U . Every point of U therefore lies in the sampled coset with probability $q^{h-\dim U}$, and its image matches the independent target with probability q^{-K} . Summing over all points gives (51).

For an ordered pair of distinct roots in one coset, write the second point as $x + y$ with $0 \neq y \in L$. The collision condition is

$$g(x + y) = g(x) \iff T_y(x) = -(g(y) - g(0)).$$

It has $\kappa(y)q^{\dim U - \text{rank } T_y}$ solutions in x . Divide the total number of ordered collisions by the number $q^{\dim U - h}q^K$ of target-coset pairs to obtain (52). $\square$

7.3 Accepted roots with full column rank

Retain U, C, g, L and the trial from Theorem 7.1, and write $g = q_2 + q_1 + q_0$ by homogeneous degree. For the full-domain derivative T_y from (50), define

$$\bar{\eta}_L^U = \sum_{0 \neq y \in L} q^{-\text{rank } T_y}. \quad (53)$$

The superscript records that the trial averages over the affine right-hand side and the L -coset. The derivative therefore has domain U , not $\ker C$. Restricting it to $\ker C$ gives a different, generally larger quantity used for fixed-fiber uniqueness, not for this experiment.

Let $X \subseteq U$ be the assignments accepted by the original verifier and put $\alpha = |X|/|U|$. A trial samples independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, and then a uniform affine coset of L inside $C^{-1}(d)$.

Theorem 7.2 (Accepted root with full column rank). *Assume $h \leq K$, $\bar{\eta}_L^U \leq \eta$, and $\alpha > \eta/(q-1)$. A trial contains an accepted root x satisfying*

$$\text{rank } D_L g(x) = h, \quad D_L g(x)[y] = T_y(x) + q_1(y),$$

with probability at least

$$p_{\text{ns}} = q^{h-K} \frac{a^2}{a + \eta}, \quad a = \alpha - \frac{\eta}{q-1}. \quad (54)$$

No independence among acceptance, singularity, the target, and the quadratic map is assumed.

Proof. Let Y count accepted roots in the sampled target-coset pair at which $D_L g$ has rank h , and let Z count all roots. For a projective direction $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$, the condition $D_L g(x)[y] = 0$ is an affine system in x whose linear part is T_y . It therefore holds on at most a $q^{-\text{rank } T_y}$ fraction of U . Union-bound over projective directions to obtain

$$\frac{|\{x : \text{rank } D_L g(x) < h\}|}{|U|} \leq \frac{1}{q-1} \sum_{0 \neq y \in L} q^{-\text{rank } T_y} \leq \frac{\eta}{q-1}.$$

Every point of U is selected by the random fiber-coset with probability $q^{h-\dim U}$ and matches the independent target with probability q^{-K} . Put $\mu = q^{h-K}$ and $a = \alpha - \eta/(q-1)$. Hence $\mathbb{E}Y \geq \mu a$. The ordered-pair collision identity gives $\mathbb{E}[Z(Z-1)] \leq \mu \eta$, and $Y(Y-1) \leq Z(Z-1)$. Therefore

$$\mathbb{E}[Y^2] = \mathbb{E}Y + \mathbb{E}[Y(Y-1)] \leq \mathbb{E}Y + \mu \eta.$$

The function $x \mapsto x^2/(x + \mu\eta)$ is increasing for $x \geq 0$. Cauchy–Schwarz therefore gives

$$\Pr[Y > 0] \geq \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y + \mu\eta} \geq \mu \frac{a^2}{a + \eta},$$

which proves (54). The denominator is strictly smaller than $1 + \eta$ whenever $a < 1$. $\square$

The following lemma bounds this spectrum in the idealized transcript model. Let $\rho_{\text{byte}} = 14/256$ be the maximum atom of one public-expansion byte reduced modulo 19.

Lemma 7.3 (Full-domain fresh-spectrum theorem). *Assume q is odd and $A = \mathbb{F}_{q^d}$. Work in the native-coordinate idealized XOF-transcript model of Section 5: the scalar entries in the fresh symmetric-matrix region are independent reductions of independent uniform bytes, each having maximum atom $\rho_{\text{byte}} = 14/256$, while all disjoint coefficient data are conditioned upon. Let L be fixed from coefficient data disjoint from that fresh region. Suppose every $0 \neq y \in L$ has nonzero projection $\bar{y}$ to a fresh A -space D , and every nonzero residual-output combination contains a nonzero coefficient $c \in \text{Sym}_{\mathbb{F}_q}^2(A)$ in at least one fresh public base form. If*

$$\dim_{\mathbb{F}_q} D = d + t,$$

then for all $0 \neq y \in L$ and $0 \neq b \in \mathbb{F}_q^K$,

$$\Pr[b \in \ker T_y^\top] \leq \rho_{\text{byte}}^t. \quad (55)$$

Consequently

$$\mathbb{E}[\bar{\eta}_L^U] \leq (q^h - 1)q^{-K} + (q^h - 1)(1 - q^{-K})\rho_{\text{byte}}^t, \quad (56)$$

$$\Pr[\bar{\eta}_L^U > (q^h - 1)q^{-K} + \varepsilon] \leq \frac{(q^h - 1)(1 - q^{-K})\rho_{\text{byte}}^t}{\varepsilon}. \quad (57)$$

For the reserved-line construction one may take $D = V' \cong A^{v-1}$ and $t = d(v-2)$. For the zero-offset construction on the full public-vinegar summand one may take $D \cong A^v$ and $t = d(v-1)$.

For fixed $y \neq 0$ and $b \neq 0$, the fresh coefficient map has rank at least t . The maximum-atom bound gives (55); expectation and Markov’s inequality give (56) and (57). The complete proof is in Section B.

For the rest of the paper, use the threshold

$$\boxed{\eta_{128}(h, K) = (q^h - 1)q^{-K} + 2^{-128}.} \quad (58)$$

This allows 2^{-128} above the exact full-rank baseline; it does not assume perfect rank.

Corollary 7.4 (Numerical spectrum envelope). *For any invoked PXL slice, let h be its base-field dimension and let t be the number of fresh scalar pivots supplied by Lemma 7.3. Substituting these two values and $\varepsilon = 2^{-128}$ into (57) gives the corresponding base-two failure exponent. The PXL ledgers record h , while the selected fresh region determines t . A fixed key can instead publish its exact projective rank histogram, which certifies (58) without the idealized XOF-transcript model.*

8 Conditional PXL recovery

The exact quotient applies to all nine parameter shapes. This section gives the audited numerical solver model for both extension degrees. The $\ell = 2$ route has exact affine coefficient support. For $\ell = 4$, we give one route supported by the graph-complement theorem and a separate cheaper route with a named structured-system premise.

8.1 The $\ell = 2$ route

For one $\ell = 2$ shape, set $h = v - 2$ and $K = 3m$. The exact accepted density on A^{v+o} is

$$\alpha_{v+o} = 19^{-(v+o)} \sum_{j=0}^{\lfloor (v+o)/4 \rfloor} \binom{v+o}{j} 18^{v+o-j}. \quad (59)$$

Put $\eta = \eta_{128}(h, K)$ and define

$$\begin{aligned} a &= \alpha_{v+o} - \frac{\eta}{18}, \\ p_{\text{reg}} &= 19^{h-K} \frac{a^2}{a + \eta}, \\ p_{\text{dir}} &= \max\{p_{\text{reg}} - 19^{-4}, 0\}. \end{aligned} \quad (60)$$

The final subtraction is a union bound with the exact nontransverse probability. It does not assume independence and is not claimed to be the exact usable trial probability.

The solver analysis uses two named premises. The ordinary MQ hypothesis H_{PXL} packages the affine semi-regularity, projected semi-regularity, and Macaulay-rank assumptions from PXL Section 4.2 [22]. The manuscript adds an instancewise premise H_{route} stating that the actual restricted matrices follow those predictions and that the added extraction procedure returns an accepted base-field root when one exists. The definitions and the complete five-component cost expression are in Section A.

Theorem 8.1 (Conditional $\ell = 2$ PXL estimates). *Consider one of the three published $\ell = 2$ shapes. Assume the exact public rank check, the spectrum inequality $\bar{\eta}_L^U \leq \eta_{128}(h, K)$ for every invoked slice, and H_{route} for every invoked restricted system. Normalize each $\mathbb{F}_{19}$ operation to the 150 Boolean gates in the released scalar multiplication circuit. This is a multiplication-based normalization, not a bound for every operation used by PXL. The conservative sum of the five components from PXL Section 4.2, target generation, and retries using (60) gives the following unit-leading gate exponents.*

ω	Level and parameters	K	h	k	D	$\log_2 W_{\text{est}}$	headroom
2.81	I, (48, 16, 2, 2)	48	46	9	12	128.833	14.167
	III, (72, 24, 2, 2)	72	70	13	17	187.030	19.970
	V, (96, 32, 2, 2)	96	94	15	23	245.042	26.958
3	I, (48, 16, 2, 2)	48	46	9	12	133.667	9.333
	III, (72, 24, 2, 2)	72	70	14	16	194.626	12.374
	V, (96, 32, 2, 2)	96	94	19	20	255.550	16.450

The headroom uses the NIST classical reference exponents 143, 207, and 272 [16]. The estimate omits setup, extraction, control, storage traffic, and final verification. If C_{aux} charges those operations per trial, Equation (77) gives the augmented expression. Every returned candidate is checked against all residual equations, the signature format rule, and the original verifier. The route therefore never returns an invalid signature, while its completeness and cost remain conditional on H_{route} .

The first theorem row uses $D = 12$ and residual matrix size $A = 99,219,059$. The unit-leading dense symbolic storage proxy is $2^{93.337}$ bits. This counts one dense symbolic array at five bits per base-field entry. It omits representation overhead, temporary arrays, and storage traffic, and is not a peak-memory bound. The corresponding residual matrix sizes and storage proxies for Levels III and V are recorded in Section A.

8.2 The support-constrained $\ell = 4$ route

For either $\ell = 4$ route, put

$$\alpha_4 = \begin{cases} (1 - 19^{-4})^{v+o}, & r = 4, \\ 1, & r > 4, \end{cases} \quad \eta = \eta_{128}(h, K), \quad p_4 = 19^{h-K} \frac{(\alpha_4 - \eta/18)^2}{\alpha_4 - \eta/18 + \eta}. \quad (61)$$

This is the accepted-root lower bound from [Theorem 7.2](#). Every $\ell = 4$ row conditions on the displayed spectrum inequality for its slice and divides by p_4 . The rectangular shapes have no square-block rejection rule, so their accepted density is one.

The supported route imposes $h - k \leq v$. It assumes the exact public checks in H_{off} , the spectrum inequality $\bar{\eta}_L^U \leq \eta_{128}(h, K)$, and the instancewise PXL premise H_{route} . The graph complement and fixed-map theorem give homogeneous channel support on the main variables. They do not give full affine coefficient support. Charging the same five PXL components, target generation, and certified retries gives the following unit-leading gate exponents.

ω	parameters	K	h	k	$h - k$	D	$\log_2 W_{\text{est}}$	$\log_2 M_{\text{sym}}^{\text{proxy}}$
2.81	(28, 5, 4, 4)	50	35	7	28	7	145.868	59.429
	(28, 4, 4, 5)	50	35	7	28	7	145.867	59.429
	(40, 7, 4, 4)	70	48	8	40	9	197.338	78.763
	(38, 5, 4, 5)	70	46	8	38	8	201.265	71.892
	(50, 9, 4, 4)	90	59	9	50	10	252.203	91.085
	(52, 6, 4, 6)	90	61	9	52	10	249.539	92.109
3	(28, 5, 4, 4)	50	35	7	28	7	148.400	59.429
	(28, 4, 4, 5)	50	35	7	28	7	148.400	59.429
	(40, 7, 4, 4)	70	49	9	40	9	201.379	79.763
	(38, 5, 4, 5)	70	46	8	38	8	205.022	71.892
	(50, 9, 4, 4)	90	60	10	50	10	257.209	92.085
	(52, 6, 4, 6)	90	61	9	52	10	254.991	92.109

For $\omega = 2.81$, the supported route exceeds the Level I reference by about 2.87 bits. It remains below the Level III and Level V reference exponents. This is the cost of enforcing the fixed-map support criterion, not a weakness in the exact 16-to-10 quotient. The official frequency of H_{off} remains unproved.

8.3 The cheaper structured $\ell = 4$ route

The cheaper route leaves more than v main variables at Level I. By [Theorem 6.5](#), it cannot have full ordinary coefficient support for any fixed realization. Full support is sufficient for the usual random MQ explanation, but it is not necessary for a structured system to have the predicted Hilbert series and Macaulay ranks. We therefore state the needed premise directly.

Definition 8.2 (Structured PXL premise H_{struct}). *For every selected $\ell = 4$ restricted system, the structured Frobenius channel family has the Hilbert series, solving degree, Macaulay ranks, and root-extraction behavior used by the five-component PXL model at the stated parameters. This premise does not assert ordinary coefficient support or a uniform MQ distribution.*

The premise is specific to this manuscript. Its form is comparable to the residual genericity assumptions used when Hashimoto-style and generalized partitioning methods replace a large underdetermined system by smaller MQ subproblems [18, 20]. The known support obstruction gives it a greater evidential burden than the ordinary PXL model. It should therefore be tested directly on the structured family.

Under H_{struct} and the same spectrum condition used in (61), the audited unit-leading gate exponents are as follows. The table does not infer ordinary MQ support from these parameters.

ω	parameters	K	h	k	D	$\log_2 W_{\text{est}}$	$\log_2 M_{\text{sym}}^{\text{proxy}}$
2.81	(28, 5, 4, 4)	50	48	10	12	134.038	95.266
	(28, 4, 4, 5)	50	48	10	12	134.038	95.266
	(40, 7, 4, 4)	70	68	12	17	182.007	135.896
	(38, 5, 4, 5)	70	68	12	17	182.006	135.896
	(50, 9, 4, 4)	90	88	15	21	230.445	172.076
	(52, 6, 4, 6)	90	88	15	21	230.444	172.076
3	(28, 5, 4, 4)	50	48	10	12	138.912	95.266
	(28, 4, 4, 5)	50	48	10	12	138.912	95.266
	(40, 7, 4, 4)	70	68	13	16	189.638	131.353
	(38, 5, 4, 5)	70	68	13	16	189.637	131.353
	(50, 9, 4, 4)	90	88	18	19	240.174	162.956
	(52, 6, 4, 6)	90	88	18	19	240.174	162.956

Every row is reproduced by the final independent ledger checker. The storage quantity is the exact five-bit dense-array proxy defined in (78); it is not total state or peak memory.

Theorem 8.3 (Conditional structured PXL estimates for all nine shapes). *Assume the exact public rank and spectrum checks for every invoked route. Assume H_{route} for the three $\ell = 2$ shapes and H_{struct} for the six $\ell = 4$ shapes. Under the five-component PXL model with $\omega = 2.81$, every displayed row lies below the NIST reference exponent for its level. Every returned candidate is checked against the complete residual system, the signature format rule, and the original verifier. Thus validity is exact, while completeness and the unit-leading gate exponents are conditional on the stated solver premises.*

9 Implementation, evidence, and reproducibility

The companion artifact provides formula checkers, one official-key reduction test, and one toy end-to-end forgery test. Table 4 states what each establishes. None executes a production-size nonlinear solver.

Both correspondence tests use a KAT-anchored Python transcription. The official-key test compares the substituted map with a separate literal evaluator but stops before the nonlinear solve. The toy forger reconstructs and serializes a signature and checks the rejection rule, but its forger and verifier share scheme helpers and do not call the official C verifier. It is therefore composition evidence rather than an independent implementation.

The release does not test H_{PXL} , H_{route} , or H_{struct} , and it does not measure the official frequency of H_{off} or the omitted auxiliary work. The numerical model charges the five PXL components, target generation, and certified retries. Control, data movement, indexing, extraction, final verification, and peak state remain outside the unit-leading gate estimate (Section A).

A production fixed-key record would need the exact preflights, a certified slice spectrum, the complete solver and filtering transcript, the reconstructed signature, and the original verifier transcript. Only an instrumented run that records the omitted work could measure total work and memory.

Table 4: Evidence shipped with the release. “Included” means that the named script and its pinned input are present, not that the headline recovery cost has been measured.

Evidence	Availability	What it establishes
Formula and ledger checks	Included	The independent PXL ledger generator and checker reproduce all 30 rows, comprising six $\ell = 2$, twelve support-constrained $\ell = 4$, and twelve structured $\ell = 4$ rows, including matrix sizes and the five-bit storage proxy.
Scalar circuit netlist	Included	The emitted 150-gate $\mathbb{F}_{19}$ multiplication netlist is correct on all 361 valid inputs. The extension-field figures are recurrence counts, not emitted netlists.
Official KAT verification and reduction	Included for one Level-I (28, 5, 4, 4) key	A Python transcription parses the compressed public key, accepts the supplied signature, rejects an altered message, and obtains quotient/affine ranks 50/30. The resulting 50-equation system in 102 variables is left unsolved.
Reduced zero-offset forgery	Included at unofficial (2, 1, 2, 2)	The complete restricted map is interpolated; its image equals the explicit rank-three quotient image, leaving one target-consistency coordinate before public slice search. The main case, eight committed fresh-key regressions, and a separate 24-forgery/100-rank stress check pass in the shared KAT-anchored Python transcription. This is toy composition evidence, not independent C-verifier or production-cost evidence.
All-shape official-key preflight census	Not included	No empirical pass probability for the reserved-line spectrum checks or H_{off} .
Production-size solver or forgery	Not included or claimed	No empirical evidence for H_{PXL} , H_{route} , H_{struct} , peak memory, or feasibility at a published shape.

10 Necessary dimension floors within the common-column model

We give two necessary output floors for redesigns that retain the common-column architecture. The first removes a square slice from every target fiber. The stronger second floor removes the corresponding zero-offset chosen-message slice. Neither floor is sufficient for security.

Proposition 10.1 (Dimensionally square common-column slice bound). *Let $A = \mathbb{F}_{q^d}$ and retain the public vinegar summand V' of base-field dimension $d(v - 1)$. Suppose an exact affine reduction has M' outputs, K' quotient quadratics, and full affine rank $M' - K'$. Then*

$$\dim_{\mathbb{F}_q}(V' \cap \ker C_{R,v}) \geq d(v - 1) - (M' - K').$$

If $M' \leq d(v - 1)$, dimensions alone leave an ordinary K' -dimensional slice carrying all K' residual equations. Equality of equation and variable counts does not establish a root, zero-dimensionality, nonsingularity, or an efficient solve. Eliminating this dimensional guarantee requires $M' > d(v - 1)$. Under the coupling $M' = dx'$, this forces $x' \geq v$ and $M' \geq dv$.

Proof. Use

$$\dim(V' \cap \ker C_{R,v}) \geq \dim V' - \text{rank } C_{R,v} = d(v - 1) - (M' - K').$$

The right side is at least K' exactly when $M' \leq d(v - 1)$. Negate and round to the next multiple of d . $\square$

Let $K_{\text{cc,ord}} = \min\{M, m_1 d^2\}$ be the pre-symmetrization ordered-label cap within the same common-column model. Retaining symmetric forms requires $m'_1 \geq \lceil K_{\text{cc,ord}} / \binom{d+1}{2} \rceil$ and $M' \geq K_{\text{cc,ord}}$. Under $M' = dx'$ and $m'_1 = \lceil x'/d \rceil$, all three necessities combine as follows.

Theorem 10.2 (Combined necessary common-column output floor). *Any redesign that retains symmetric public forms, the public vinegar count, the public-vinegar affine geometry above, full affine rank $M' - K'$, and the Version 2.3 output/base-form coupling, while both restoring $K_{\text{cc,ord}}$ and eliminating the dimensionally guaranteed fresh slice, must satisfy*

$$x' \geq \max \left\{ v, \left\lceil \frac{K_{\text{cc,ord}}}{d} \right\rceil, d \left(\left\lceil \frac{K_{\text{cc,ord}}}{\binom{d+1}{2}} \right\rceil - 1 \right) + 1 \right\}, \quad M' = dx'. \quad (62)$$

Across the nine official rows, this requires 45.00%–60.71% more outputs ([Theorem 10.4](#)).

Proof. The second and third terms in (62) are the two ordered-cap requirements under $m'_1 = \lceil x'/d \rceil$. The term v is [Proposition 10.1](#). Their maximum is therefore necessary. Substitution gives the affine-fiber column in the comparison below. $\square$

Zero-offset targets. The preceding floor assumes a full-rank affine lift. A zero-offset attacker can instead filter chosen-message targets through the quadratic image and use the full common-column domain. Removing this square slice requires a stronger floor.

Proposition 10.3 (Zero-offset dimensionally square restriction). *Consider any symmetric redesign for which the zero-offset common-column relation with $R_0 = 1$ is defined. Let Q_R be its quadratic output matrix and put $\rho_Q = \text{rank } Q_R$. Public row reduction gives an invertible output change under which the zero-offset verifier is equivalent to*

$$g_{\rho_Q}(u) = z \in \mathbb{F}_q^{\rho_Q}, \quad t_{\text{con}} = 0 \in \mathbb{F}_q^{M' - \rho_Q}, \quad (63)$$

where g_{ρ_Q} consists of ρ_Q independent quadratic output combinations. For an exactly uniform target, the consistency event in (63) has probability $q^{-(M' - \rho_Q)}$ and, conditioned on it, z is uniform in $\mathbb{F}_q^{\rho_Q}$.

With zero offsets no reserved line is needed, so use the full public-vinegar space $V_{\text{vin}} \cong A^v$, of base-field dimension dv . If

$$\rho_Q \leq dv, \quad (64)$$

then dimensions alone leave an ordinary ρ_Q -dimensional slice $L \subseteq V_{\text{vin}}$ carrying all ρ_Q residual equations: it has ρ_Q equations in ρ_Q variables. This dimension equality alone proves neither root existence nor efficient recovery. Consequently, ruling out this chosen-message dimensional configuration requires

$$\rho_Q > dv. \quad (65)$$

Proof. With zero offsets, the substituted verifier is $Q_R z_{\text{sym}}(u) = t$ and contains no affine or constant term. Choose a row basis of Q_R and extend it to an invertible basis change on the M' output coordinates. The first ρ_Q transformed rows are the independent residual quadratics g_{ρ_Q} ; the remaining rows vanish identically on the quadratic image and therefore impose the consistency equations in (63). An invertible linear transformation sends a uniform target to independent uniform residual and consistency coordinates, proving the distribution statement. Under (64), choose any ρ_Q -dimensional linear subspace of V_{vin} and restrict all ρ_Q residual equations to an affine coset of it. This gives the asserted dimensionally square restriction. $\square$

Symmetry bounds the actual quotient rank by

$$\rho_Q \leq \min \left\{ M', m'_1 \binom{d+1}{2} \right\}. \quad (66)$$

Hence making (65) even possible forces both terms on the right of (66) above dv .

Theorem 10.4 (Output floor eliminating the zero-offset dimension guarantee). *Retain symmetry, the public vinegar count, and the Version 2.3 coupling*

$$M' = dx', \quad m'_1 = \left\lceil \frac{x'}{d} \right\rceil.$$

Put

$$C_d = \binom{d+1}{2}, \quad m_{\text{sq}} = \left\lfloor \frac{dv}{C_d} \right\rfloor + 1.$$

A necessary condition for eliminating the zero-offset chosen-message dimension guarantee is

$$x' \geq \max \{v + 1, d(m_{\text{sq}} - 1) + 1\}, \quad M' = dx'. \quad (67)$$

If the redesign also restores the common-column ordered-label cap $K_{\text{cc,ord}}$, the right side of (67) must additionally be maximized with

$$\left\lceil \frac{K_{\text{cc,ord}}}{d} \right\rceil, \quad d \left(\left\lceil \frac{K_{\text{cc,ord}}}{C_d} \right\rceil - 1 \right) + 1.$$

For the nine official rows, the zero-offset condition dominates the ordered-cap terms. The two repair floors compare as follows.

Level	parameters	current M	necessary outputs M'	
			affine fiber	zero offset
<i>I</i>	(28, 5, 4, 4)	80	116 (45.00%)	180 (125.00%)
<i>I</i>	(48, 16, 2, 2)	64	96 (50.00%)	130 (103.125%)
<i>I</i>	(28, 4, 4, 5)	80	116 (45.00%)	180 (125.00%)
<i>III</i>	(40, 7, 4, 4)	112	180 (60.71%)	260 (132.14%)
<i>III</i>	(72, 24, 2, 2)	96	144 (50.00%)	194 (102.08%)
<i>III</i>	(38, 5, 4, 5)	100	152 (52.00%)	244 (144.00%)
<i>V</i>	(50, 9, 4, 4)	144	228 (58.33%)	324 (125.00%)
<i>V</i>	(96, 32, 2, 2)	128	192 (50.00%)	258 (101.5625%)
<i>V</i>	(52, 6, 4, 6)	144	228 (58.33%)	324 (125.00%)

The zero-offset floor requires 101.5625%–144.00% more outputs.

Proof. Condition (65) and the two caps in (66) require $dx' > dv$ and $m'_1 C_d > dv$. The first inequality is equivalent to $x' \geq v + 1$. The second requires $m'_1 \geq m_{\text{sq}}$, and the least integer x' with $\lceil x'/d \rceil \geq m_{\text{sq}}$ is $d(m_{\text{sq}} - 1) + 1$. This proves (67). The two extra ordered-cap terms are the same feature and output necessities used in Theorem 10.2. Direct substitution gives the table. $\square$

The stronger result is a dimension-only theorem. A redesign below the table can retain the square system and rely on the consistency factor $q^{M' - \rho_Q}$, its solving cost, or a changed target interface. The floor applies only if the redesign instead removes the square slice while preserving the common-column architecture and parameter coupling.

11 Conclusion

The common-column restriction exposes the central weakness of the $q = 19$ Version 2.3 construction. Symmetry collapses its restricted quadratic coordinates to a symmetric-square quotient. Conditional on the stated public ranks, an invertible output change separates this quotient from the affine and target-consistency constraints. The result is a quadratic system on a public affine space. Every affine restriction preserves the verifier equations, the rejection rule, and final verification. This statement concerns the restricted verifier, not the unrestricted matrix-valued map.

All nine published shapes admit the exact quotient and affine restriction (Corollary 4.4). For $\ell = 2$, transverse slices have full affine coefficient support. The resulting PXL rows are unit-leading gate exponents under the ordinary MQ assumptions and the instancewise route premise (Theorem 8.1). For $\ell = 4$, the stronger 16-to-10 quotient, the fixed-map criterion, and its zero-offset graph realization are exact for homogeneous channels. Positive-rank offset compatibility remains conditional on H_{off} , whose frequency under the official distribution is unproved. Under the instancewise PXL premise, the support-constrained route crosses Levels III and V but not Level I. Under the separate structured PXL premise, the estimates cross all six $\ell = 4$ rows (Theorems 6.5 and 8.3). The artifact contains no production-size solve or forgery (Table 4).

The repair bounds apply only to designs that retain the common-column architecture and current parameter coupling. They show that restoring the ordered-label count does not by itself remove the stronger zero-offset slice (Theorems 10.2 and 10.4).

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A Complete PXL model and numerical details

This appendix states the solver premises and all charged terms behind [Theorem 8.1](#). PXL first eliminates part of a Macaulay matrix over a polynomial ring in variables that are fixed later. It then evaluates those variables and performs the remaining linear algebra over the base field [\[22\]](#).

Definition A.1 (Ordinary MQ regularity hypothesis H_{PXL}). *Let K affine quadratic equations in h variables be given. Choose k variables for polynomial preprocessing and let the remaining $h - k$ variables be the main variables. For a degree bound D , the hypothesis $H_{\text{PXL}}(K, h, k, D)$ consists of the following statements.*

- (i) *For every assignment to the k fixed variables, the specialized system has at most one root over the algebraic closure, counted with multiplicity, and its highest-degree parts are semi-regular.*
- (ii) *The homogeneous quadratic parts in the main variables are semi-regular, so their Hilbert series gives the predicted residual matrix size.*
- (iii) *The selected Macaulay rows have the predicted rank, and polynomial preprocessing leaves a matrix of the predicted size.*
- (iv) *Degree D suffices for the second linearization step to produce the predicted elimination data.*

These assumptions package the affine semi-regularity, projected semi-regularity, and Macaulay-rank assumptions used in PXL Section 4.2. They are not implied by coefficient support and are not proved for the quotient systems in this paper.

Definition A.2 (Instancewise route premise H_{route}). *Fix a public key, retained target, and selected affine slice, and let f be the resulting restricted system. In the $\ell = 2$ route this is the system $f_{a,z}$ in [\(43\)](#). The premise $H_{\text{route}}(f; k, D)$ states that its PXL matrices have the degrees, dimensions, and ranks specified by $H_{\text{PXL}}(K, h, k, D)$. It also states that, when the system has an accepted base-field root with full column rank, the manuscript’s specialization and extraction procedure returns at least one such root. The extraction statement is added by this manuscript and is not part of the published PXL assumptions.*

For fixed (K, h, k, D) , the predicted residual matrix size is

$$A = \sum_{d=0}^D \max \left\{ [t^d](1-t)^{K-h+k}(1+t)^K, 0 \right\}. \quad (68)$$

The model chooses the least $D \geq 2$ such that

$$[t^D](1-t)^{K-h+k-1}(1+t)^K \leq 1. \quad (69)$$

This is a degree prediction, not a solving-degree theorem for the structured quotient system.

Put $B = \binom{h-k+D}{D}$. Section 4.2 of PXL gives five cost components. We charge all five and aggregate them by addition. This sum is our conservative use of those components. The preprocessing costs are

$$C_1 = B^\omega, \quad (70)$$

$$C_2 = k^2(h-k)B^2, \quad (71)$$

$$C_3 = k^2AB \binom{h+D}{D}. \quad (72)$$

After preprocessing, evaluation and the second linearization cost

$$C_{\text{fix}} = 19^k A^2 \binom{k+D}{D}, \quad (73)$$

$$C_{\text{lin}} = 19^k A^\omega. \quad (74)$$

The first three terms are paid once. The last two charge all 19^k assignments to the fixed variables.

Define

$$C_{\text{PXL}} = C_1 + C_2 + C_3 + C_{\text{fix}} + C_{\text{lin}}, \quad C_{\text{target}} = 2^{32} 19^m. \quad (75)$$

Here m is the target-consistency codimension of the route. The factor 2^{32} is a stipulated unit-leading model convention per retained target. It is not a measured cost or a circuit-level bound. Multiplying the field-operation count by the 150 gates in the released $\mathbb{F}_{19}$ multiplication circuit gives the multiplication-based gate normalization

$$W_{\text{est}} = \frac{150 C_{\text{PXL}} + C_{\text{target}}}{p_{\text{dir}}}. \quad (76)$$

The factor 150 is not a bound for every operation used by the solver. If C_{aux} charges setup, extraction, control, storage traffic, and final verification for one trial, the augmented expression is

$$W_{\text{aug}} = \frac{150 C_{\text{PXL}} + C_{\text{target}} + C_{\text{aux}}}{p_{\text{dir}}}. \quad (77)$$

Both formulas place target generation inside the retry numerator. The root theorem samples a fresh retained target and affine coset in every trial.

The independently checked parameters and matrix sizes are as follows. The search is performed separately for each value of ω .

ω	Level	k	D	A	$\log_2 W_{\text{est}}$	$\log_2 M_{\text{sym}}^{\text{proxy}}$	reference exponent
2.81	I	9	12	99,219,059	128.833	93.337	143
	III	13	17	2,710,186,080,528	187.030	137.856	207
	V	15	23	547,365,920,911,569,552	245.042	186.447	272
3	I	9	12	99,219,059	133.667	93.337	143
	III	14	16	962,082,316,942	194.626	133.178	207
	V	19	20	9,230,629,773,094,340	255.550	172.754	272

Here

$$M_{\text{sym}}^{\text{proxy}} = 5B^2 \binom{k+D}{D} \text{ bits}. \quad (78)$$

This is a unit-leading dense proxy for one symbolic preprocessing array. It is not total state or peak memory.

For Level I, the five charged terms have the following base-two logarithms. This row makes the aggregation in (75) reproducible.

ω	$\log_2 C_1$	$\log_2 C_2$	$\log_2 C_3$	$\log_2 C_{\text{fix}}$	$\log_2 C_{\text{lin}}$
2.81	102.354	84.399	109.027	109.525	112.877
3	109.275	84.399	109.027	109.525	117.924

B Proofs for slice preparation and spectrum bounds

This subsection collects the coefficient-level proofs used in [Sections 5](#) and [7](#). The corresponding statements and proof ideas remain in the main body.

Proof of Theorem 5.1. The family in (25) has $m_1 d^2$ members. It is stable under right multiplication by A because, for every $\zeta \in A$,

$$(w^\top \Sigma_\xi P_i \Sigma_\eta) \Sigma_\zeta = w^\top \Sigma_\xi P_i \Sigma_{\eta\zeta},$$

which is another member of the span. Every offset cross term contains one copy of w . If w occurs on the right, symmetry and commutativity move it to the left. Thus every row of $F_{R,v}$ lies in U_w , and right stability puts every orbit row there as well.

The complete right orbit in (26) makes its row space an A -vector space, so $d \mid \rho_F$. Its first block is $F_{R,v}$, hence its kernel kills every offset-linear feature. If x is in the kernel and $\zeta \in A$, each $S^a \zeta$ is an $\mathbb{F}_q$ -linear combination of the chosen basis powers. The same orbit rows therefore kill $\Sigma_\zeta x$, proving that the kernel is A -linear. Rank-nullity over A gives (27). $\square$

Proof of Lemma 5.2. Every structured offset lies in W , so each offset-linear coefficient contains one copy of w . Symmetry moves that copy to the left when needed. The coefficient therefore reads only native matrix entries with at least one index in W . Right multiplication by A preserves this coordinate block, as do the deterministic orbit and slice constructions.

The native $V' \times V'$ entries come from disjoint XOF symbols and remain unread. They are therefore independent of the selected slice in the stated model. This argument must stay in the native coordinates because the biased byte-reduction distribution is not invariant under an arbitrary A -linear basis change. $\square$

Proof of Lemma 5.3. Choose t pivot columns. Condition on all nonpivot variables. Expose the pivot variables in echelon order; at each step at most one value can satisfy the next independent equation, and that value has probability at most ρ_{byte} . Multiplication gives the claim. $\square$

Proof of Theorem 5.4. First consider failure of the affine source rank. If $\text{rank } C_{R,v} < M - K$, some nonzero $\lambda \in \mathcal{Q}_R^\vee$ annihilates $C_{R,v}$. In particular, $(\lambda \circ C_{R,v})|_{V'} = 0$. For fixed $[\lambda]$, this is an affine system of rank at least $d(v-1)$ in the independent native variables $\mathbf{X}_{W,V'}$; [Lemma 5.3](#) bounds its probability by $\rho_{\text{byte}}^{d(v-1)}$. Scalar multiples define the same event, and $\mathbb{P}(\mathcal{Q}_R^\vee)(\mathbb{F}_q)$ has $(q^{M-K} - 1)/(q - 1)$ points. A union bound proves (29).

Next consider failure of the projection to V' . By [Theorem 5.1](#), $H_{R,v}^F$ is A -linear. A nonzero intersection with $W \oplus O_{\text{pub}}$ must therefore contain an A -line $\mathfrak{l}$. For $\mathfrak{l} = W$, the coefficient-rank hypothesis and [Lemma 5.3](#) give probability at most $\rho_{\text{byte}}^{m_1 \binom{d+1}{2}}$. For every other A -line the analogous bound is $\rho_{\text{byte}}^{m_1 d^2}$. The $(1+o)$ -dimensional A -space $W \oplus O_{\text{pub}}$ contains $(|A|^{1+o} - 1)/(|A| - 1)$ lines, exactly one of which is W ; a union bound gives (30). Finally, the kernel of the projection $H_{R,v}^F \rightarrow V'$ is precisely $H_{R,v}^F \cap (W \oplus O_{\text{pub}})$.

A final union bound combines the source-rank and projection failures without assuming independence. Direct substitution gives the stated numerical bound. $\square$

Proof of Lemma 7.3. Fix $y \neq 0$ and $b \neq 0$. Choose a fresh public base form whose corresponding symmetric-square coefficient c is nonzero. Writing $c = \sum_\nu a_\nu \odot b_\nu$, the fresh symmetric-matrix coefficient in the scalar derivative $b \cdot T_y(x)$ is

$$\Xi_{c,\bar{y}}(x) = \sum_\nu ((a_\nu \bar{y}) \odot (b_\nu x) + (b_\nu \bar{y}) \odot (a_\nu x)).$$

We first bound the kernel of this coefficient map. If $x \notin A\bar{y}$, project onto the cross summand between the disjoint lines $A\bar{y}$ and Ax . After identifying both lines with A , this projection is the ordinary symmetrization of c . It is nonzero because symmetrization is injective on $\text{Sym}^2(A)$ in odd characteristic. Hence the kernel is contained in $A\bar{y}$, and the coefficient map has rank at least $\dim_{\mathbb{F}_q} D - d = t$.

The event $b \in \ker T_y^\top$ forces a rank- t affine system in independent fresh symbols. Conditioning on the nonpivot symbols and applying the maximum-atom bound to the pivots proves (55).

Finally, the dual-kernel identity gives

$$q^{-\text{rank } T_y} = q^{-K} |\ker T_y^\top| = q^{-K} \left(1 + \sum_{b \neq 0} \mathbf{1}_{\{b \in \ker T_y^\top\}} \right).$$

Its expectation is at most $q^{-K} + (1 - q^{-K})\rho_{\text{byte}}^t$. Summing over $y \neq 0$ proves (56). The excess above the baseline $(q^h - 1)q^{-K}$ is nonnegative, so Markov's inequality proves (57). The two stated choices of D have dimensions $d(v - 1)$ and dv . $\square$

An unused fixed-skew format check. A deterministic alternative is also available: for the official matrix

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \quad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets with $(v_1)_{b,0} - (v_0)_{b,1} = 1$ in every block. Then each block has fixed nonzero skew 1, so every root passes the rejection rule. This fixed-skew route is a useful separate algebraic check but is not needed for the reported recovery theorem.

C A finite fixed-key spectrum-certificate format

The idealized XOF-transcript theorem is not required for a statement about a concrete key. This appendix gives a finite exact certificate format for the aggregate spectrum. No such production-key spectrum certificate is shipped in the release.

Put $N_U = \dim_{\mathbb{F}_q} U$. Choose a basis $y_0, \dots, y_{h-1}$ of a fixed slice L and write

$$T(Y) = \sum_{i=0}^{h-1} Y_i T_{y_i} \in \mathbb{F}_q[Y_0, \dots, Y_{h-1}]^{K \times N_U}.$$

For $0 \leq j < h$, use the disjoint normalized projective chart $Y_0 = \dots = Y_{j-1} = 0$, $Y_j = 1$, and put $R_j = \mathbb{F}_q[Y_{j+1}, \dots, Y_{h-1}]$. Let $T^{(j)}$ be the resulting matrix. For $t \geq 0$, define

$$I_{j,t} = \left\langle Y_i^q - Y_i \ (i > j), \quad \text{all } (t+1) \times (t+1) \text{ minors of } T^{(j)} \right\rangle \subseteq R_j. \quad (79)$$

Put $n_{j,\leq t} = \dim_{\mathbb{F}_q}(R_j/I_{j,t})$, $n_{j,\leq -1} = 0$, and

$$N_t = \sum_{j=0}^{h-1} (n_{j,\leq t} - n_{j,\leq t-1}). \quad (80)$$

Proposition C.1 (Projective rank-histogram certificate). *The integer N_t is exactly the number of projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ for which $\text{rank } T_y = t$. Consequently*

$$\bar{\eta}_L^U = (q-1) \sum_t N_t q^{-t}. \quad (81)$$

Reduced Gröbner bases with their standard monomial sets, or polynomial reduction transcripts proving the unit-ideal and quotient-dimension claims, form machine-checkable fixed-key certificates.

Proof. The normalized charts contain exactly one representative of every projective $\mathbb{F}_q$ -direction. The field equations identify

$$R_j / \langle Y_i^q - Y_i : i > j \rangle \cong \prod_{a \in \mathbb{F}_q^{h-j-1}} \mathbb{F}_q$$

by evaluation. Every ideal in this product of fields is radical, and its quotient dimension is the number of coordinates on which all generators vanish. The minors in (79) vanish exactly when $\text{rank } T^{(j)}(a) \leq t$. Thus $n_{j, \leq t}$ counts rank-at-most- t points in chart j , and differencing gives (80). Every projective direction has $q - 1$ nonzero scalar representatives and rank is invariant under base-field scaling, proving (81). $\square$

A stronger certificate that $\text{rank } T_y \geq t_0 + 1$ for every nonzero y consists of unit-ideal transcripts for all I_{j, t_0} . The complete histogram is more flexible: a small exceptional low-rank locus can be charged exactly instead of controlling the whole pencil by its minimum rank.

D Exact $\ell = 4$ PXL ledger rows

The following tables record the exact inputs and outputs in the final PXL ledger. The generator and checker verify the formulas independently. The support-constrained rows condition on $\mathbf{H}_{\text{off}}$ and $\mathbf{H}_{\text{route}}$. The structured rows condition on $\mathbf{H}_{\text{struct}}$ and do not assert ordinary coefficient support. Together with the $\ell = 2$ table in Section A, these are the 30 rows checked by the artifact.

Table 5: Exact support-constrained $\ell = 4$ PXL ledger rows.

ω	parameters	h	k	D	A	$\log_2 W_{\text{est}}$	$\log_2 M_{\text{sym}}^{\text{proxy}}$
2.81	(28, 5, 4, 4)	35	7	7	60,179	145.867751	59.429
	(28, 4, 4, 5)	35	7	7	60,179	145.867386	59.429
	(40, 7, 4, 4)	48	8	9	4,927,544	197.338396	78.763
	(38, 5, 4, 5)	46	8	8	1,606,800	201.265361	71.892
	(50, 9, 4, 4)	59	9	10	108,000,036	252.202579	91.085
	(52, 6, 4, 6)	61	9	10	461,010,165	249.538960	92.109
3	(28, 5, 4, 4)	35	7	7	60,179	148.400390	59.429
	(28, 4, 4, 5)	35	7	7	60,179	148.400025	59.429
	(40, 7, 4, 4)	49	9	9	4,927,544	201.379285	79.763
	(38, 5, 4, 5)	46	8	8	1,606,800	205.022213	71.892
	(50, 9, 4, 4)	60	10	10	108,000,036	257.208548	92.085
	(52, 6, 4, 6)	61	9	10	461,010,165	254.991164	92.109

Table 6: Exact structured $\ell = 4$ PXL ledger rows.

ω	parameters	h	k	D	A	$\log_2 W_{\text{est}}$	$\log_2 M_{\text{sym}}^{\text{proxy}}$
2.81	(28, 5, 4, 4)	48	10	12	125,101,086	134.037961	95.266
	(28, 4, 4, 5)	48	10	12	125,101,086	134.037595	95.266
	(40, 7, 4, 4)	68	12	17	2,232,564,665,436	182.006674	135.896
	(38, 5, 4, 5)	68	12	17	2,232,564,665,436	182.006152	135.896
	(50, 9, 4, 4)	88	15	21	14,990,431,104,743,028	230.444951	172.076
	(52, 6, 4, 6)	88	15	21	14,990,431,104,743,028	230.444296	172.076
3	(28, 5, 4, 4)	48	10	12	125,101,086	138.911992	95.266
	(28, 4, 4, 5)	48	10	12	125,101,086	138.911626	95.266
	(40, 7, 4, 4)	68	13	16	811,505,434,656	189.638021	131.353
	(38, 5, 4, 5)	68	13	16	811,505,434,656	189.637499	131.353
	(50, 9, 4, 4)	88	18	19	706,353,332,229,810	240.174424	162.956
	(52, 6, 4, 6)	88	18	19	706,353,332,229,810	240.173769	162.956

E Arithmetic-circuit certificate details

The scalar multiplier consumes two little-endian five-bit canonical integers in $\{0, \dots, 18\}$ and returns the canonical five-bit representation of their product modulo 19. Its 150 gates are two-input AND, XOR, and XNOR gates with free constants, wires, fan-out, and structural sharing. The released netlist is tested on all 361 valid input pairs. The SHA-256 digest of the canonical compact JSON payload containing exactly `ninputs`, `gates`, and `outputs` is

`e1708a32c1475ad14295c2b34fed976fb09577b127810d29bf237db18aabc131`

for the emitted balanced-netlist representation. This is an internal payload identifier, not the bitwise digest of the surrounding ledger file.

The tower recurrences are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2 + 1), \quad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2 - (1 + u)).$$

Three-product Karatsuba at each level gives separately checked recurrence counts 692 and 2628. The artifact verifies the tower identities, the basis determinant, and the recurrence arithmetic; it does not emit or exhaustively test complete extension-field netlists. These extension-field counts are not used in the PXL tables.

The scalar multiplication circuit supplies the multiplication-based normalization used in the PXL tables. The circuits are not lower bounds and do not price the whole solver. Additions, inversions, factorization, bulk evaluation/interpolation transforms, filtering, control, indexing, communication, and memory are outside the reported W_{est} and belong to C_{aux} if charged.

F Reproduction and package contents

The source package is self-contained. Its root contains the complete paper source, bibliography, compiled PDF, build instructions, and hashes. The `artifact/` directory contains:

- the final PXL generator and pinned JSON ledger for all 30 rows, comprising six $\ell = 2$, twelve support-constrained $\ell = 4$, and twelve structured $\ell = 4$ rows;
- an aggregate verifier that reruns the PXL checker and the retained formula, target-sampling, dimension, and repair-floor checks;
- the complete 150-gate scalar multiplier netlist and tower checks;
- a public-key correspondence harness for one official Level-I KAT; and
- a non-planted zero-offset end-to-end composition test at the unofficial shape $(2, 1, 2, 2)$, with fresh-key forgery and rank stress checks.

A clean reproduction is:

```
sha256sum -c SHA256SUMS
make regenerate
make verify
make
```

The PXL ledger records H_{off} only for the support-constrained $\ell = 4$ route and H_{struct} only for the structured route. It does not assign either premise an empirical probability. The checkers are deterministic, require Python 3.10 or newer plus NumPy for the two correspondence tests, and do not require network access. The paper build uses only the files in the package. The supplied `SHA256SUMS` records the release-critical files listed in the manifest. Generated TeX build files are excluded.